Predicting College Student Performance Using Machine Learning Techniques

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University of the Incarnate Word

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Open University Learning Analytics Dataset

About Open University

- One of the largest universities in Europe with enrollment of 200,000
- Headquarters based in Milton Keynes, Buckinghamshire, England
- Popular for its flexible distance learning programs

About Open University Learning Analytics Dataset

- Contains student demographic, registration, assessment and daily interactions with courses in OU's Virtual Learning Environment from the years 2013 and 2014
- The data is anonymized
- This dataset contains 22 courses with 32,593 students

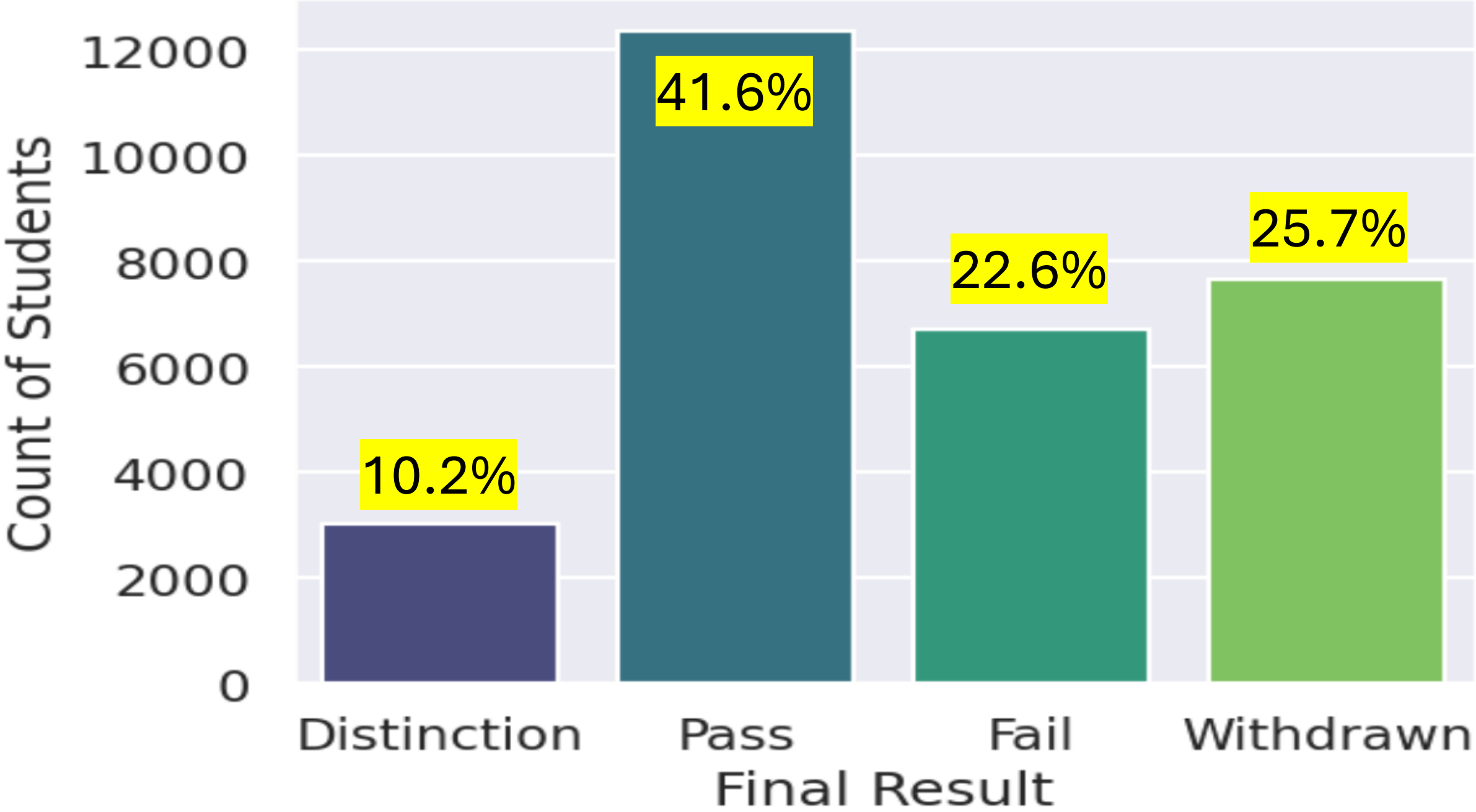
Exploratory Data Analysis

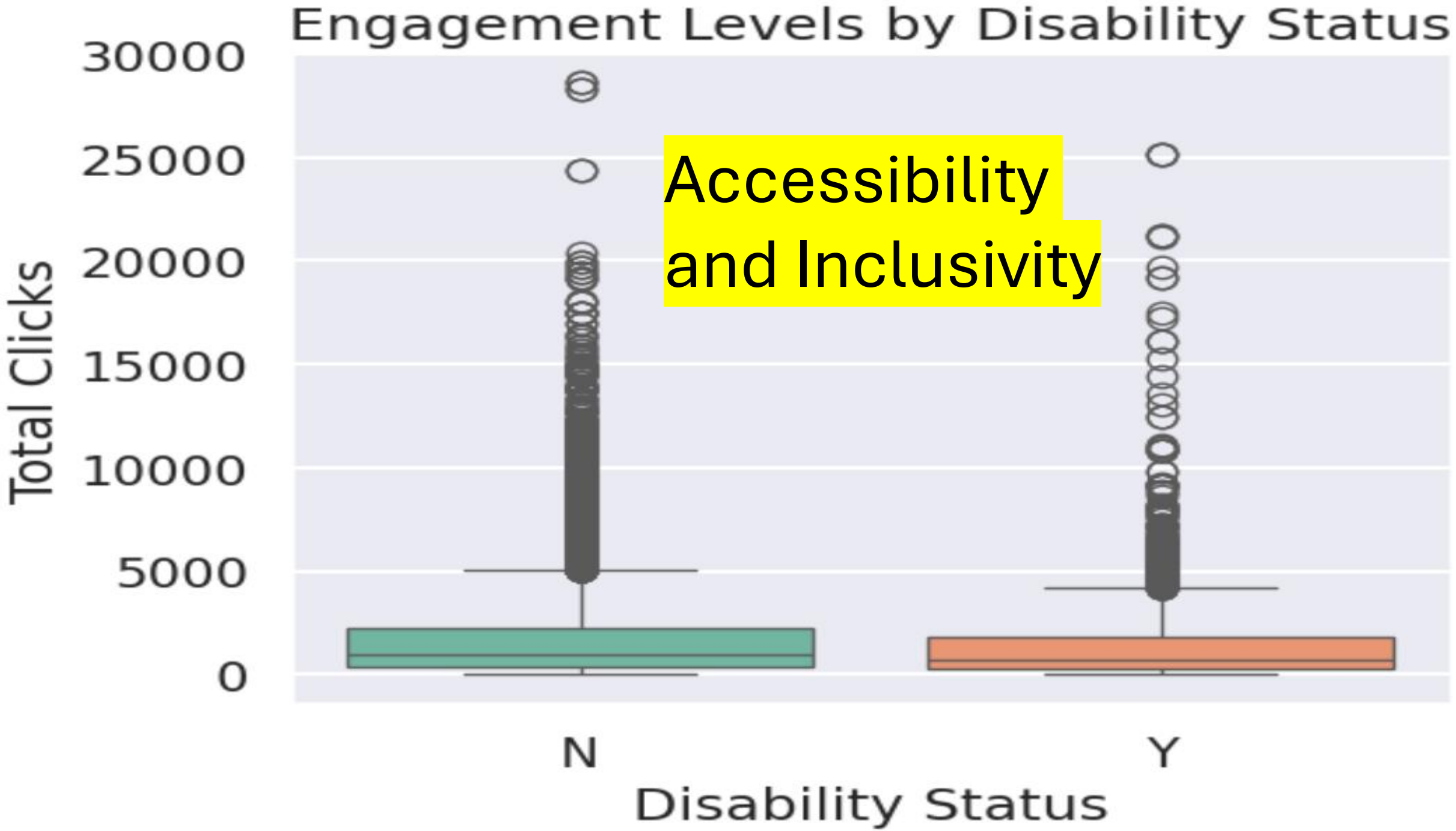
Objectives

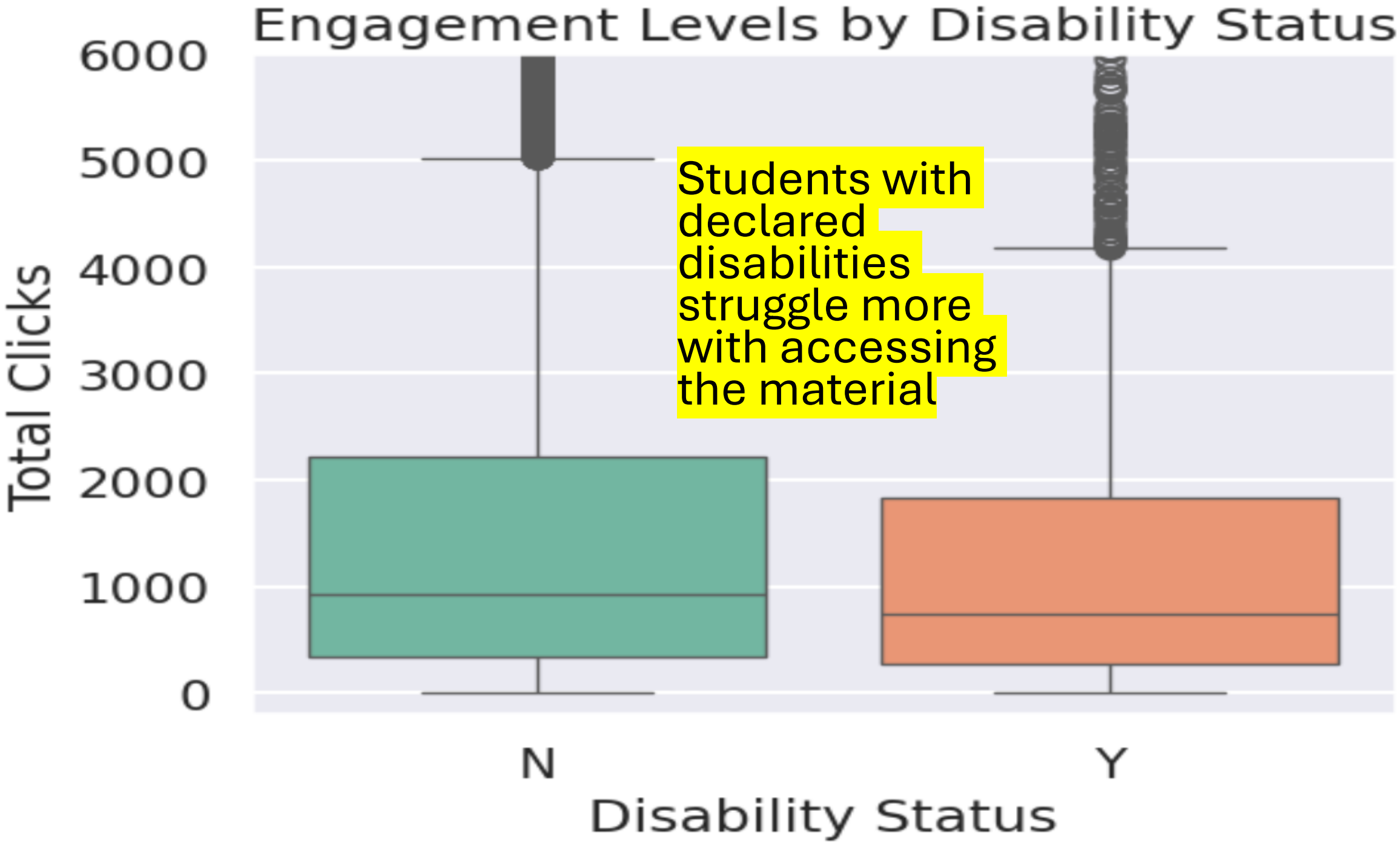
Using Exploratory Data Analysis, our goal is to:

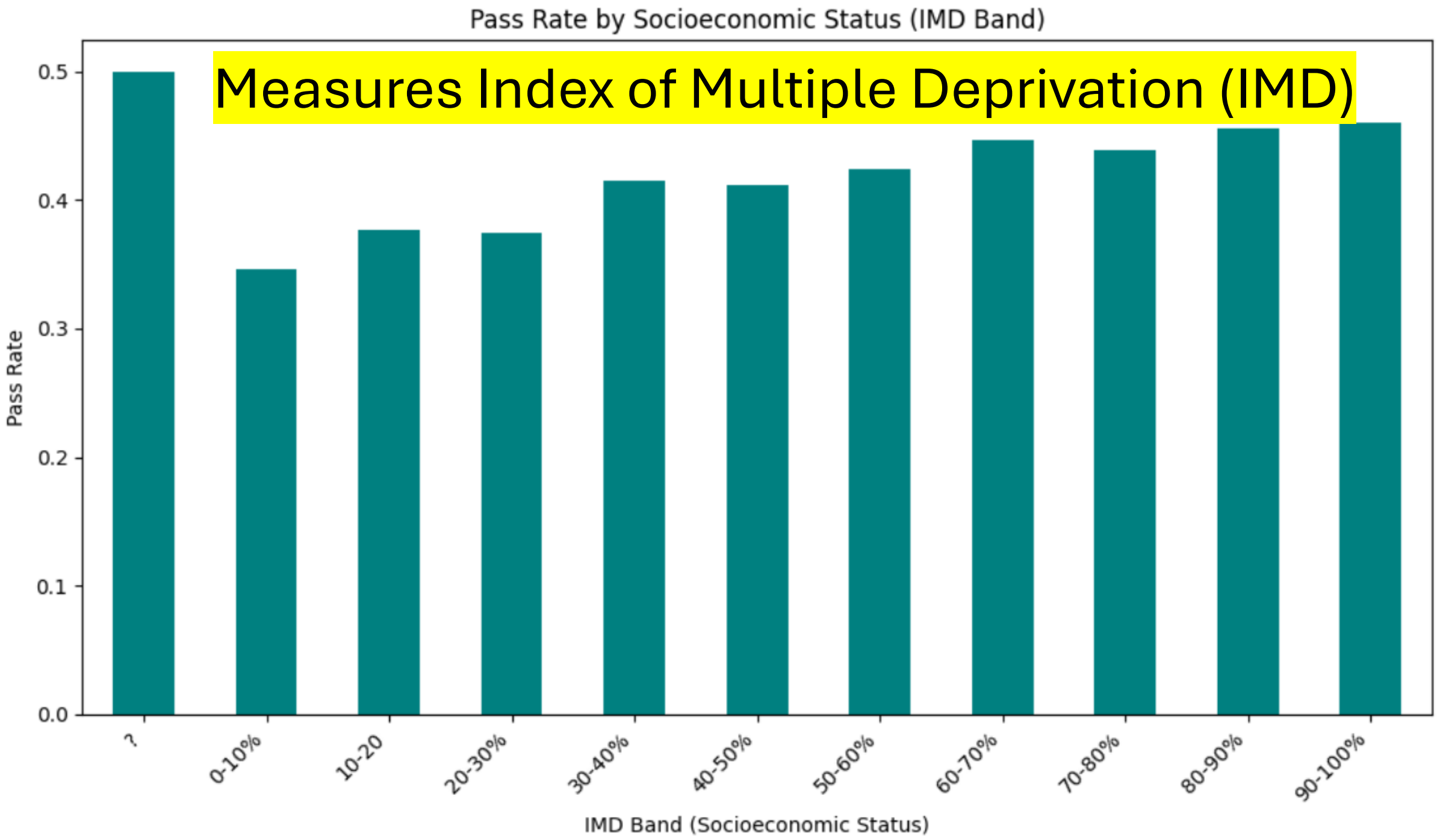
- Identify the course outcomes, including the withdrawal and failure rates.
- Determine if the course is accessible and inclusive, regardless of student background or disability.
- Identify study habits that lead to course success.
- Identify efficient and inefficient course navigation that may impact student success

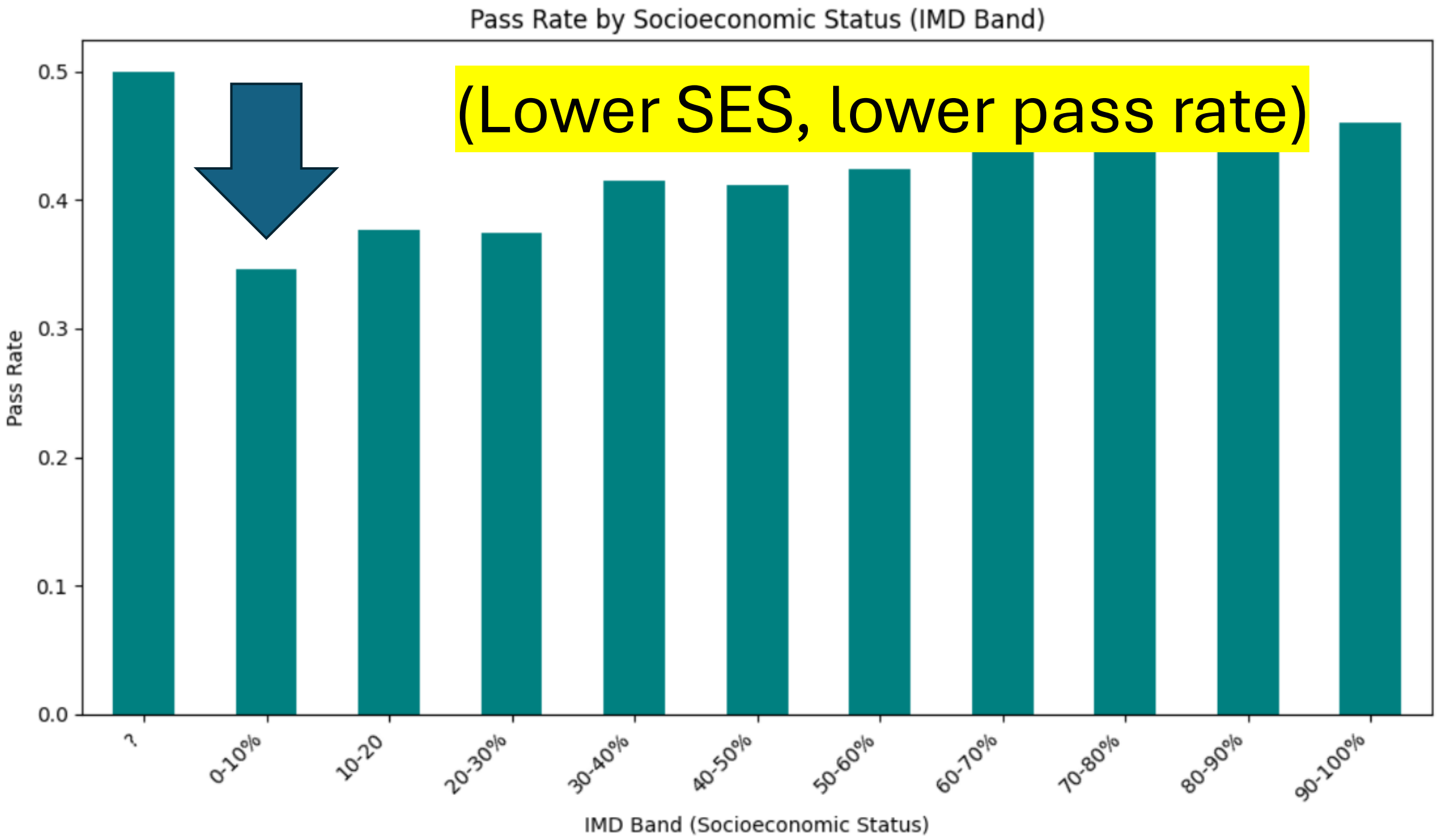
Course Grade Outcomes of Students

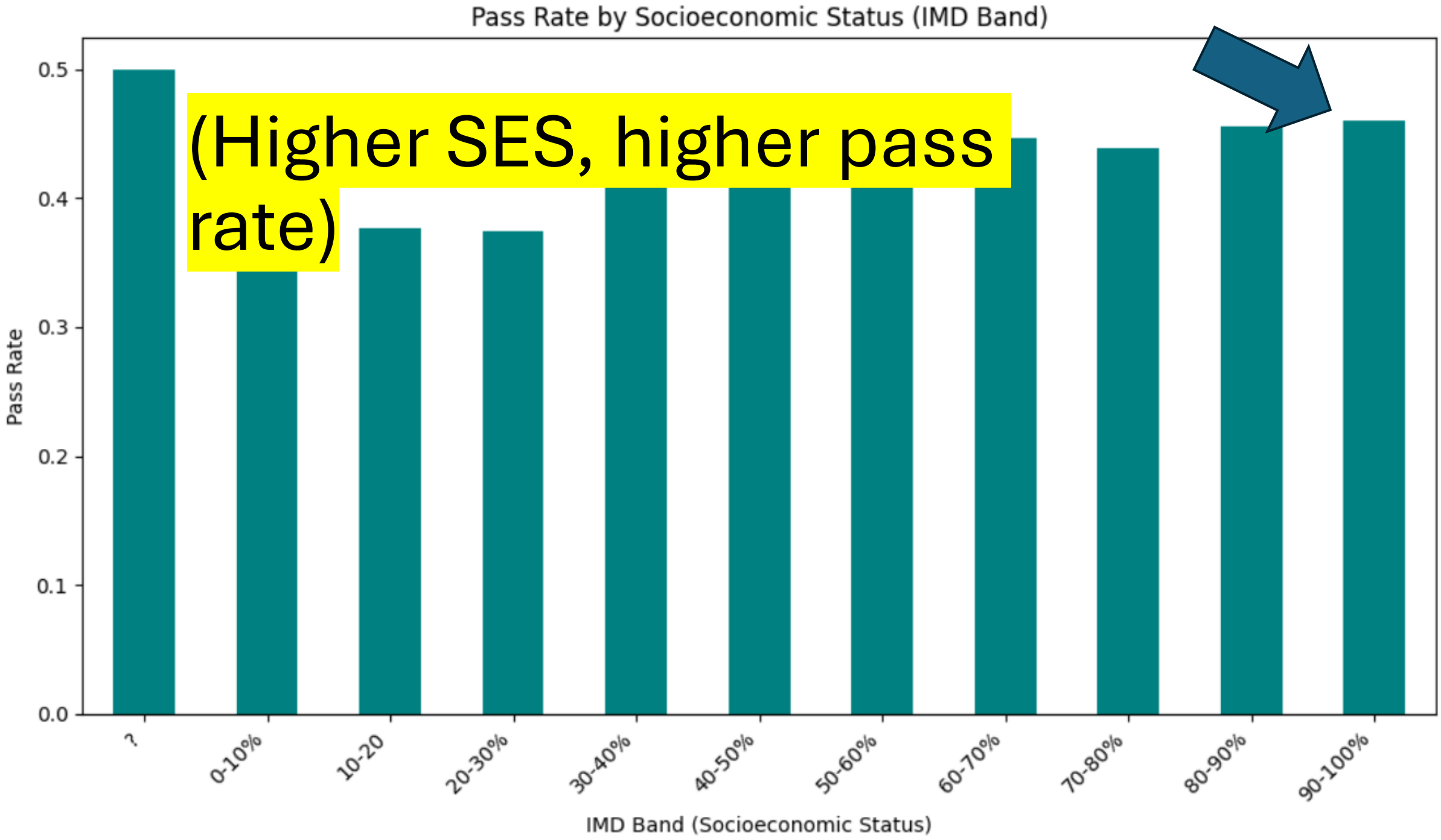


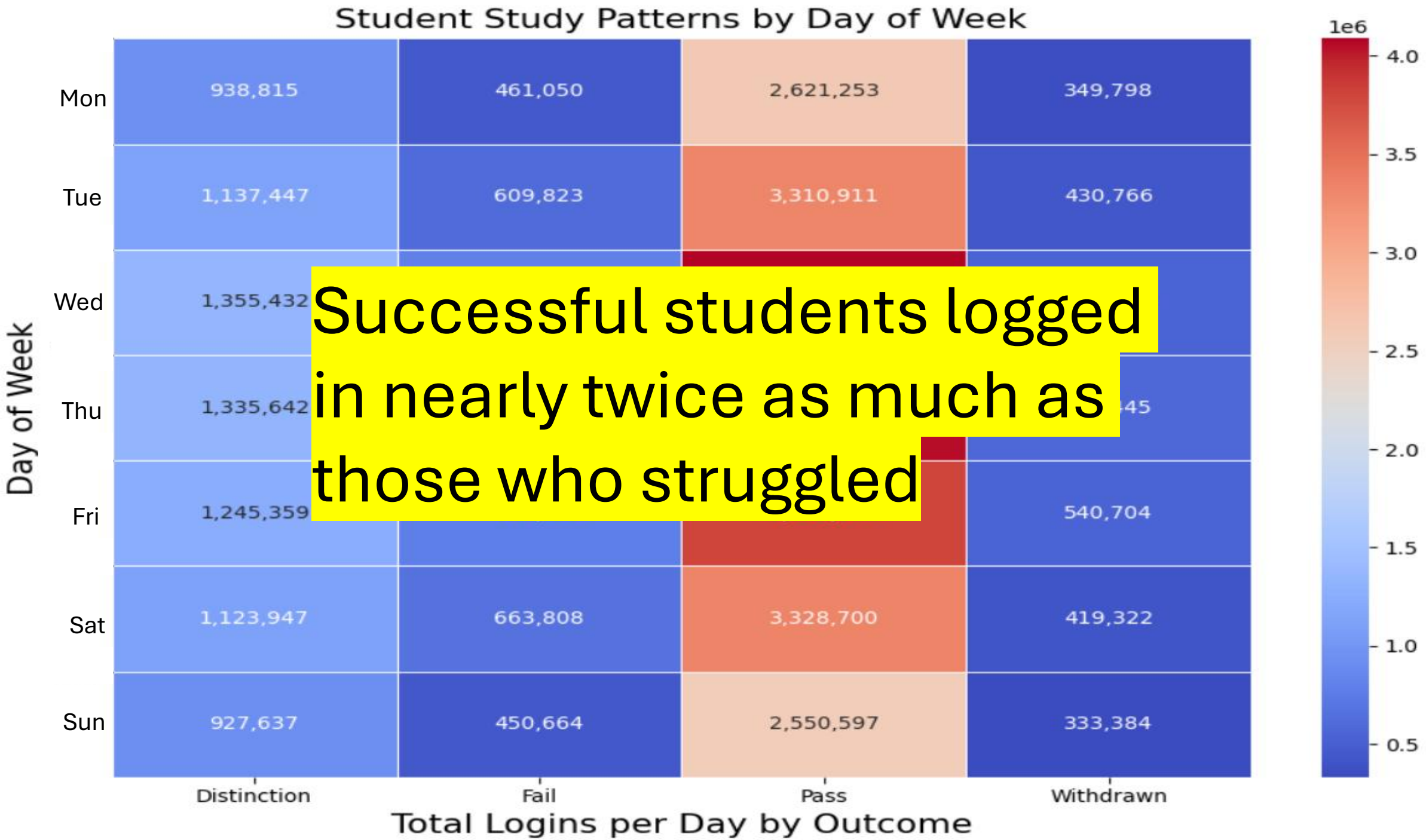

Engagement Levels by Disability Status
Accessibility
and Inclusivity
Total Clicks
30000
25000
20000
15000
10000
5000
0
N
Y
Disability Status


Engagement Levels by Disability Status
Total Clicks
6000
5000
4000
3000
2000
1000
0
N
Y
Disability Status
Students with declared disabilities struggle more with accessing the material

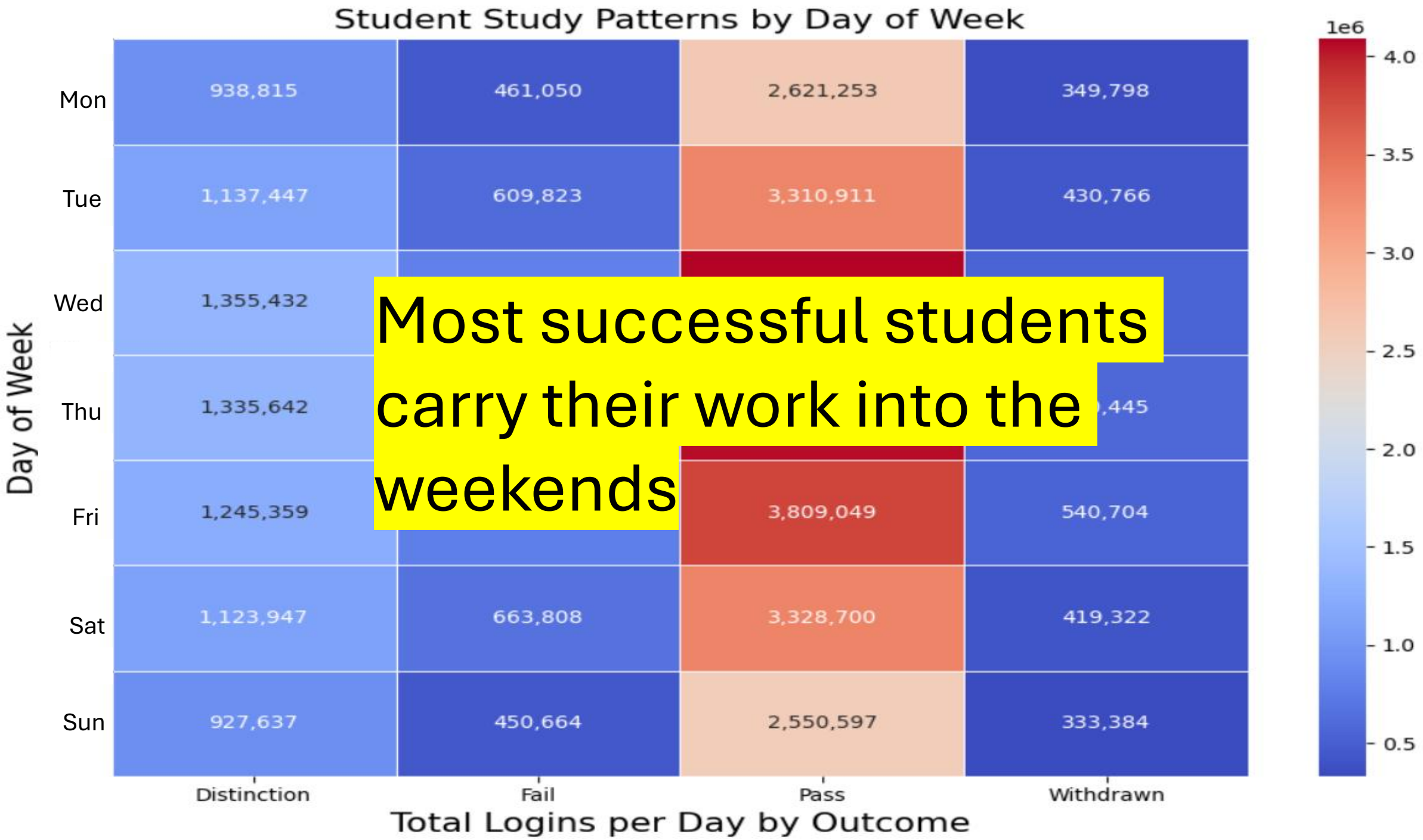

Pass Rate by Socioeconomic Status (IMD Band)
Measures Index of Multiple Deprivation (IMD)
Pass Rate
0.0
0.1
0.2
0.3
0.4
0.5
?
0-10%
10-20
20-30%
30-40%
40-50%
50-60%
60-70%
70-80%
80-90%
90-100%
IMD Band (Socioeconomic Status)


Pass Rate by Socioeconomic Status (IMD Band)
(Lower SES, lower pass rate)
Pass Rate
0.0
0.1
0.2
0.3
0.4
0.5
?
0-10%
10-20
20-30%
30-40%
40-50%
50-60%
60-70%
70-80%
80-90%
90-100%
IMD Band (Socioeconomic Status)

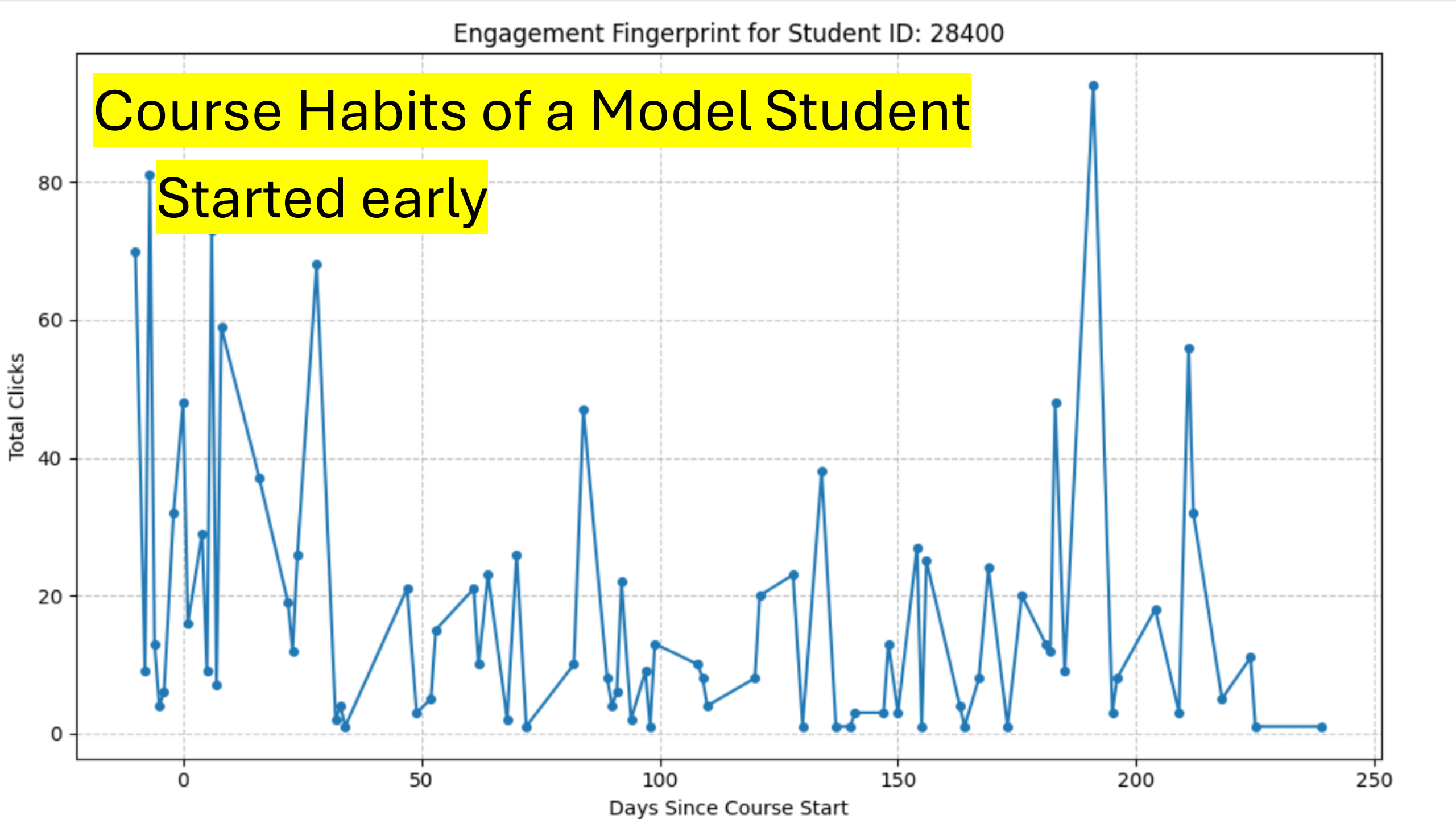

Pass Rate by Socioeconomic Status (IMD Band)
(Higher SES, higher pass rate)
Pass Rate
0.0
0.1
0.2
0.3
0.4
0.5
?
0-10%
10-20
20-30%
30-40%
40-50%
50-60%
60-70%
70-80%
80-90%
90-100%
IMD Band (Socioeconomic Status)

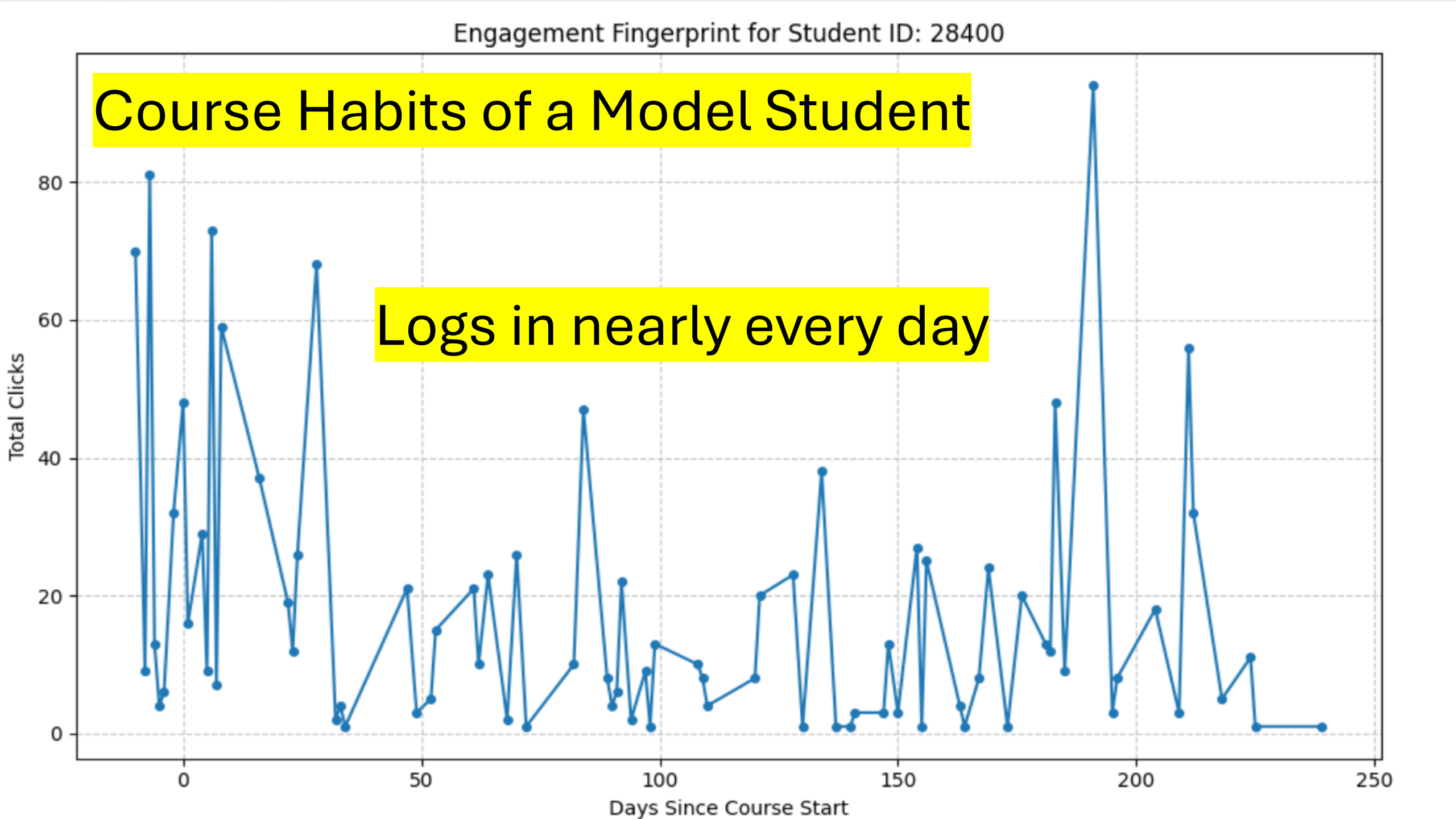


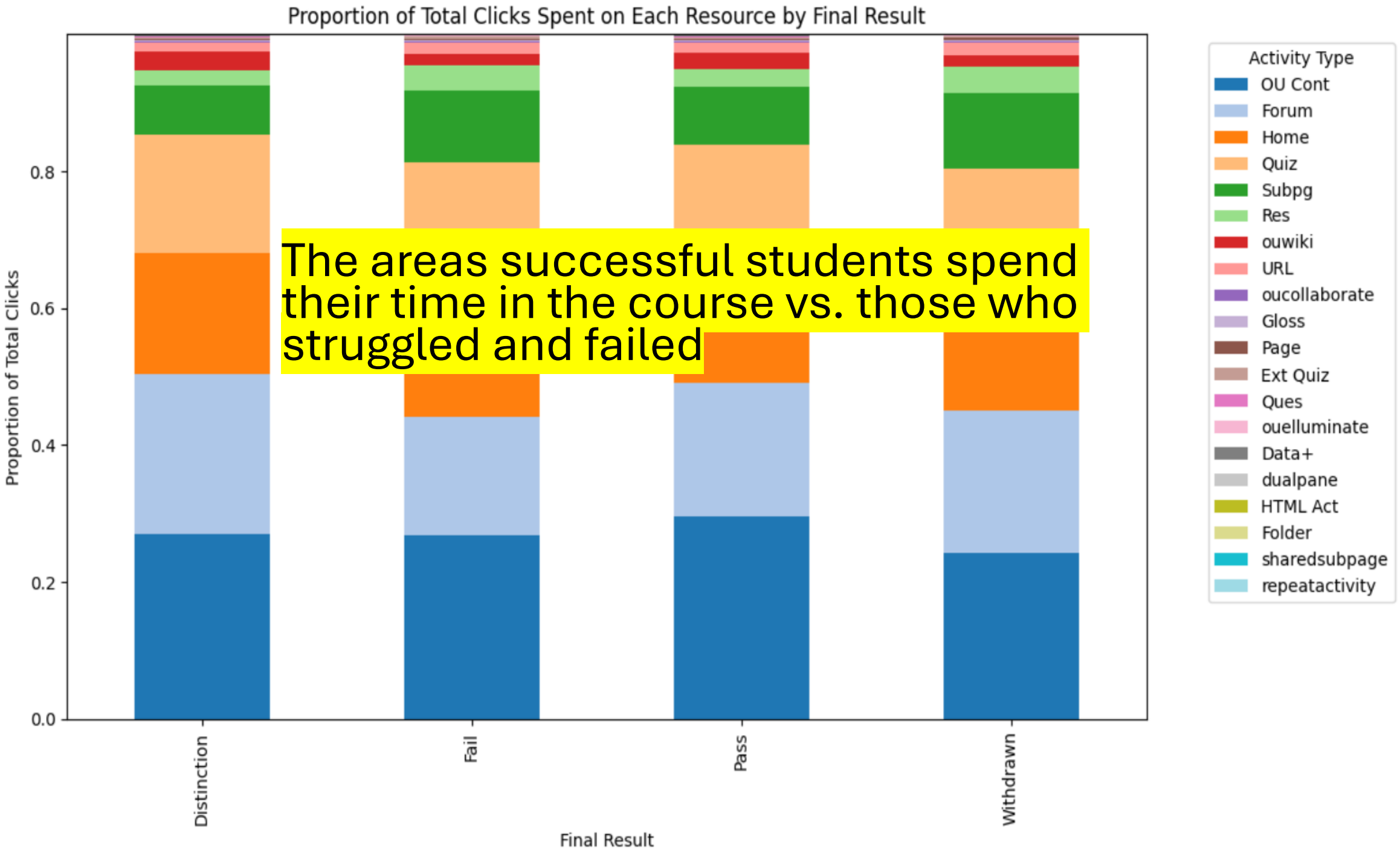
Successful students logged in nearly twice as much as those who struggled



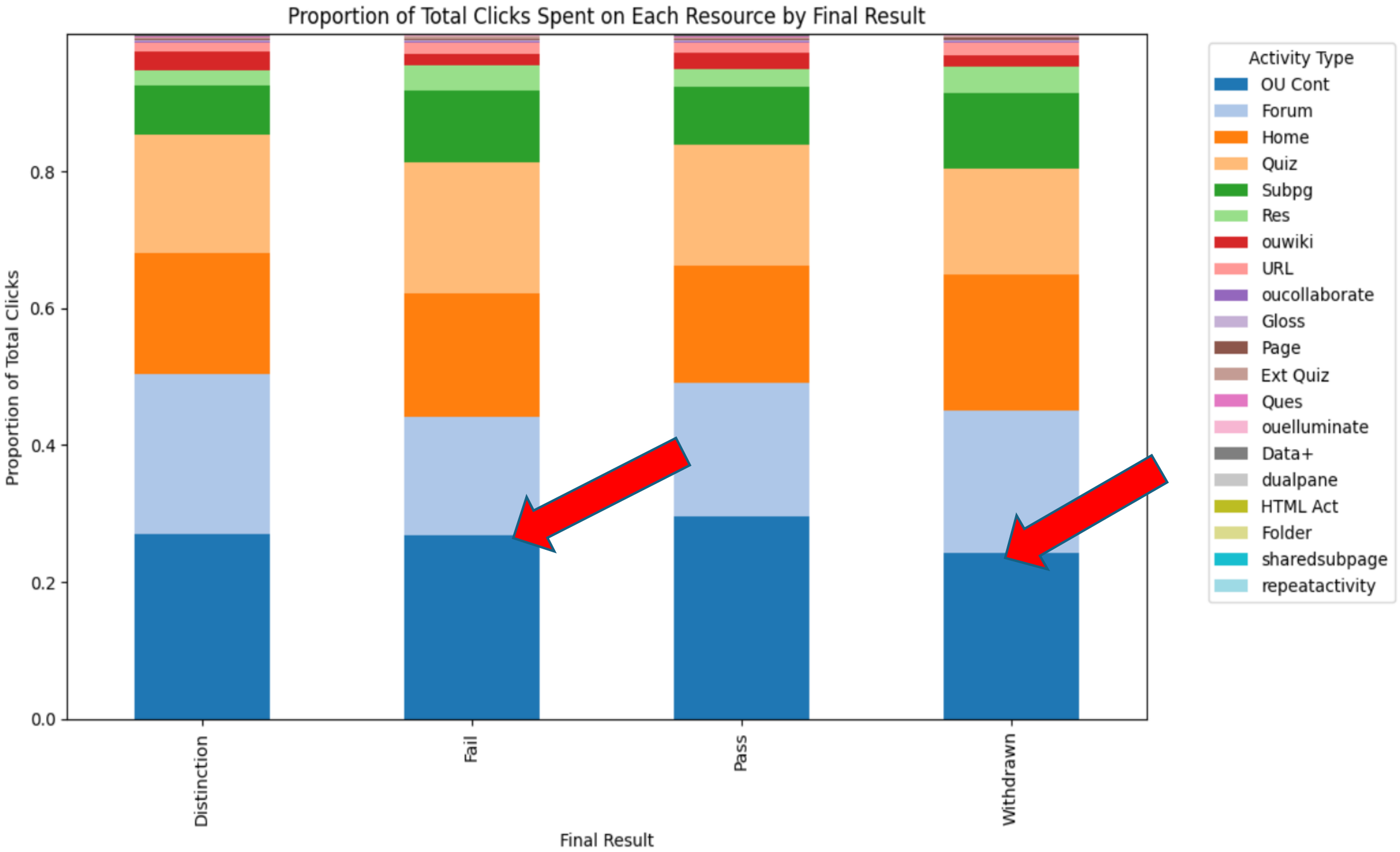
Most successful students carry their work into the weekends

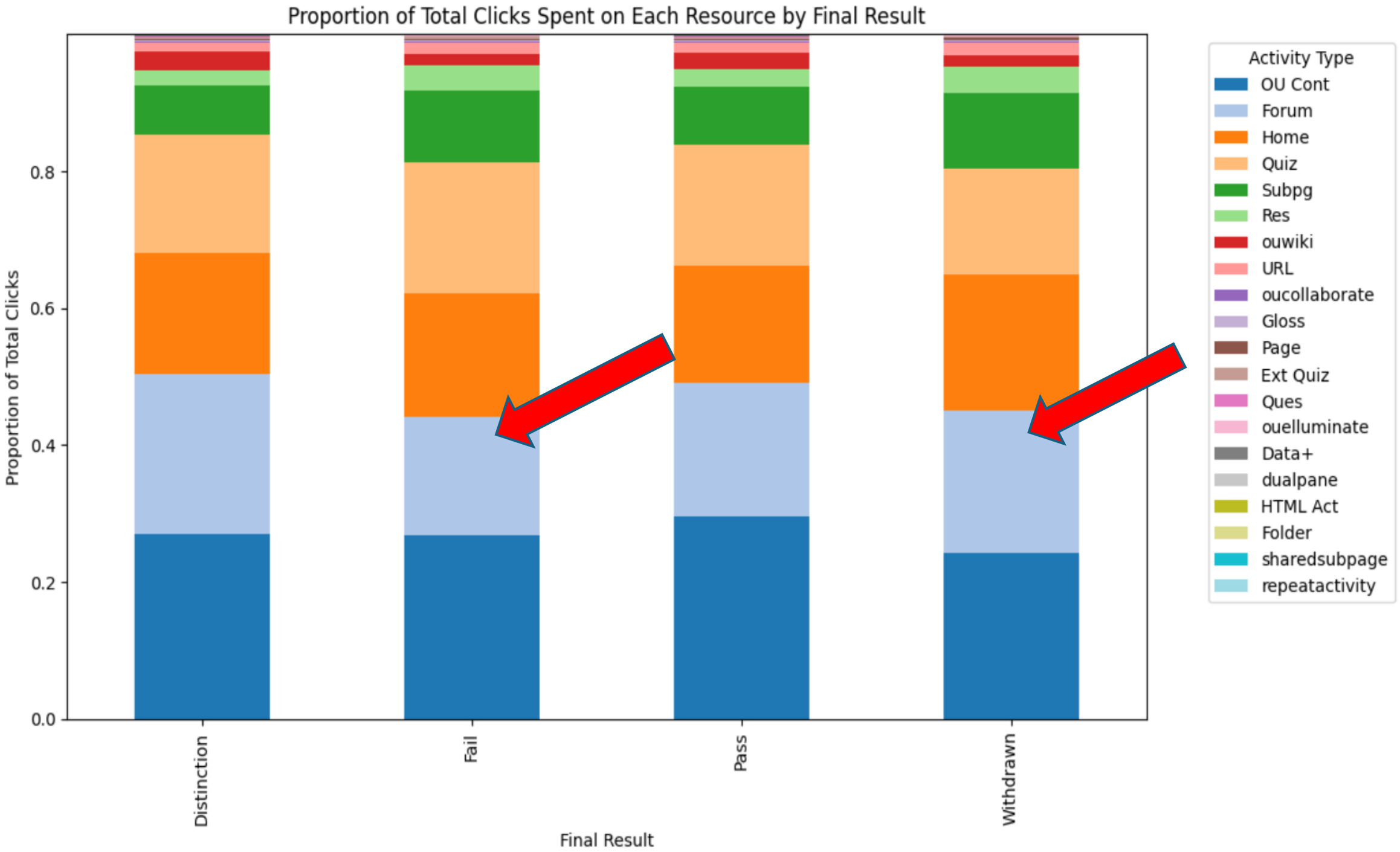

Engagement Fingerprint for Student ID: 28400
Course Habits of a Model Student
Started early
Total Clicks
0
20
40
60
80
0
50
100
150
200
250
Days Since Course Start


Engagement Fingerprint for Student ID: 28400
Course Habits of a Model Student
Logs in nearly every day
Total Clicks
0
20
40
60
80
0
50
100
150
200
250
Days Since Course Start



The areas successful students spend their time in the course vs. those who struggled and failed


Proportion of Total Clicks Spent on Each Resource by Final Result
Proportion of Total Clicks
0.0
0.2
0.4
0.6
0.8
Distinction
Fail
Pass
Withdrawn
Final Result
Activity Type
OU Cont
Forum
Home
Quiz
Subpg
Res
ouwiki
URL
oucollaborate
Gloss
Page
Ext Quiz
Ques
ouelluminate
Data+
dualpane
HTML Act
Folder
sharedsubpage
repeatactivity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0.2
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dualpane
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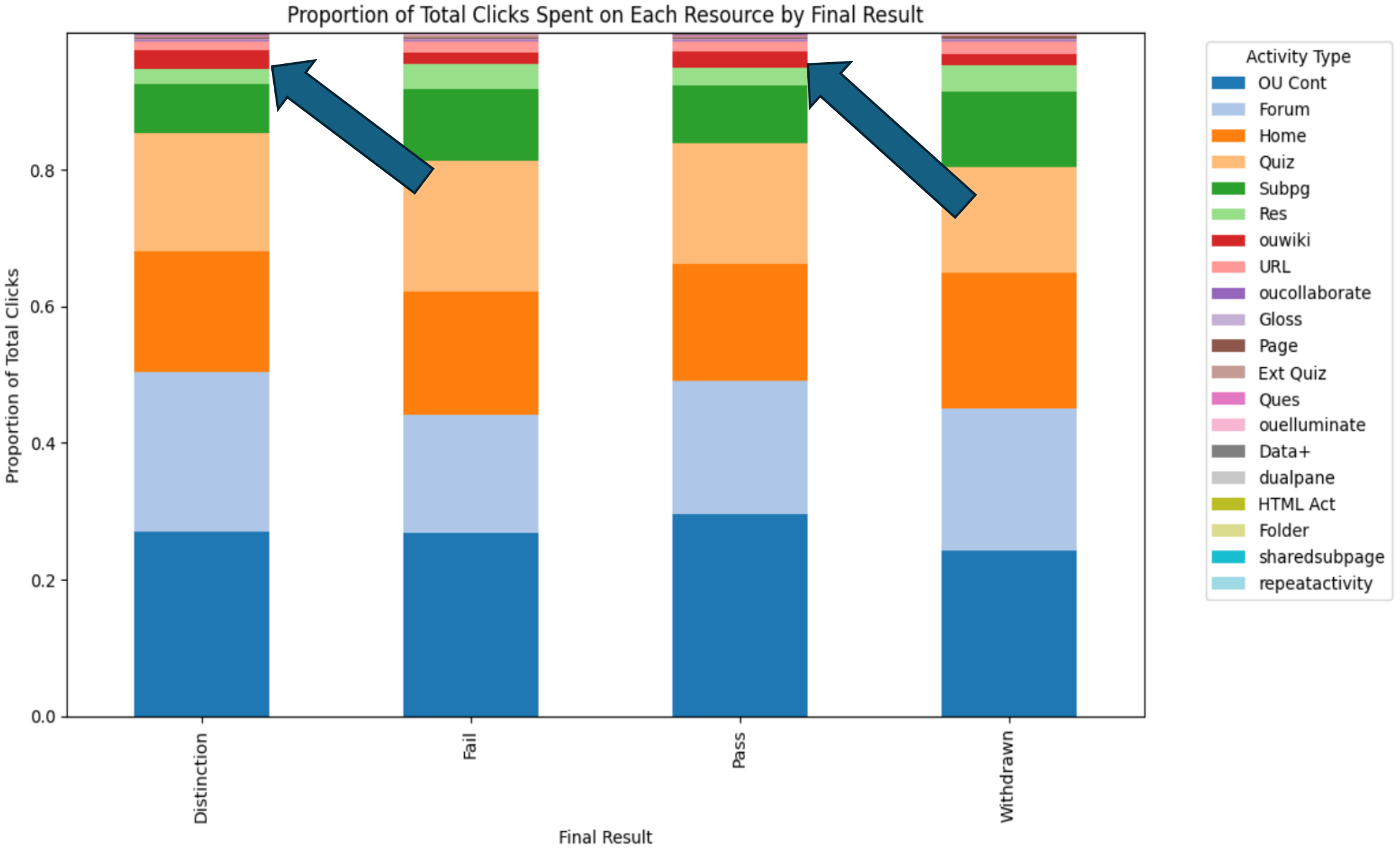
Folder
sharedsubpage
repeatactivity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Folder
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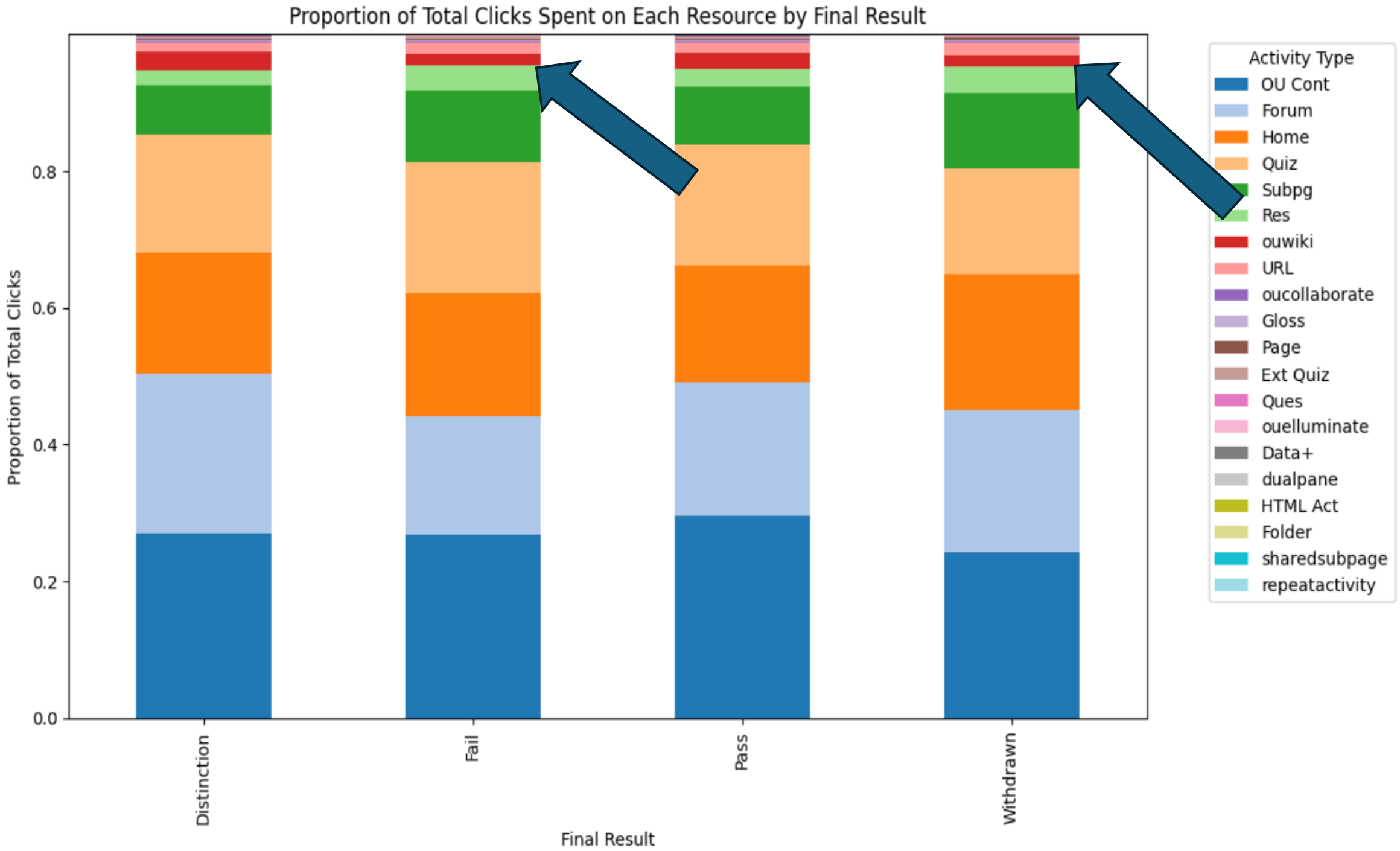
repeatactivity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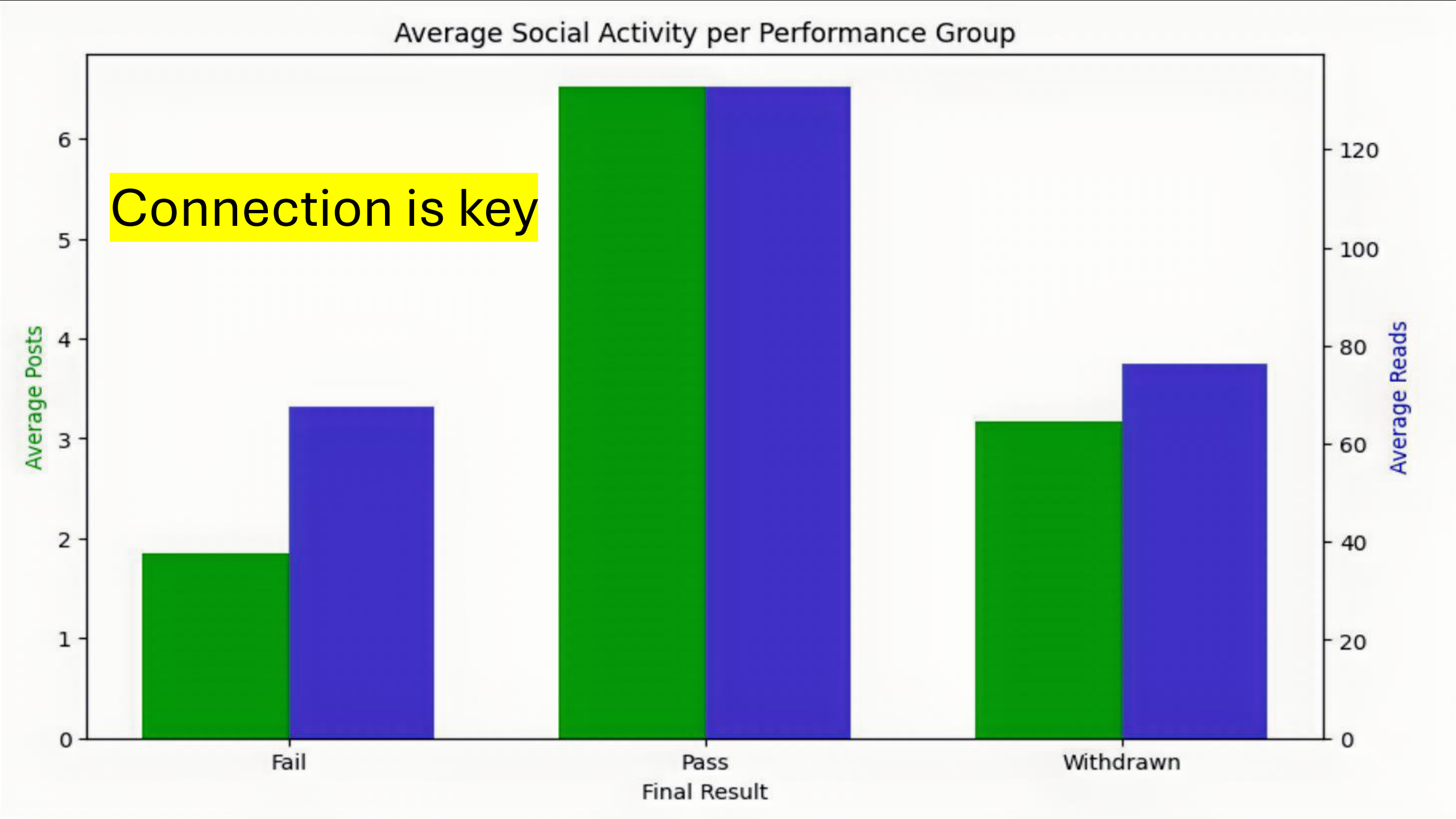
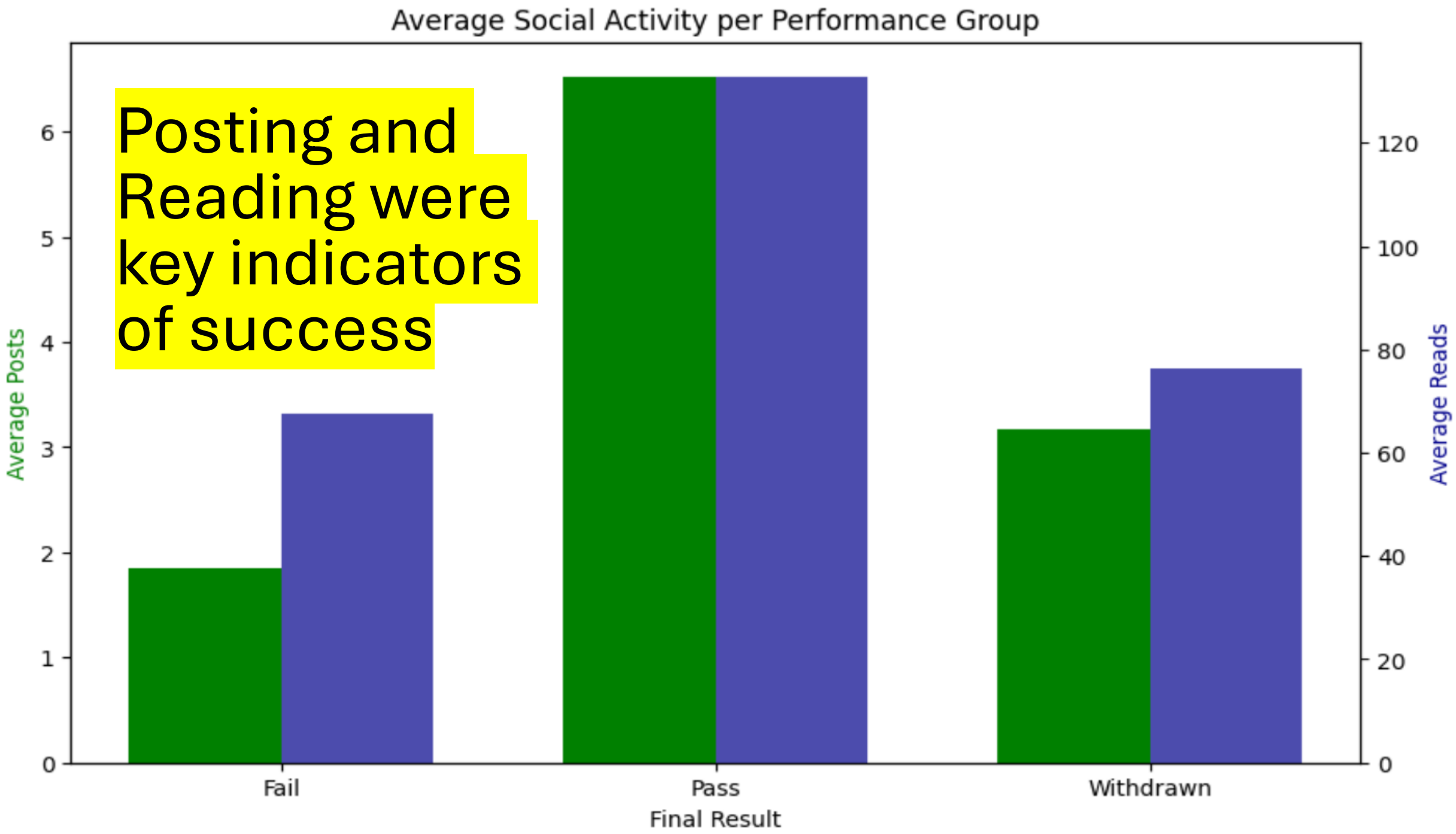
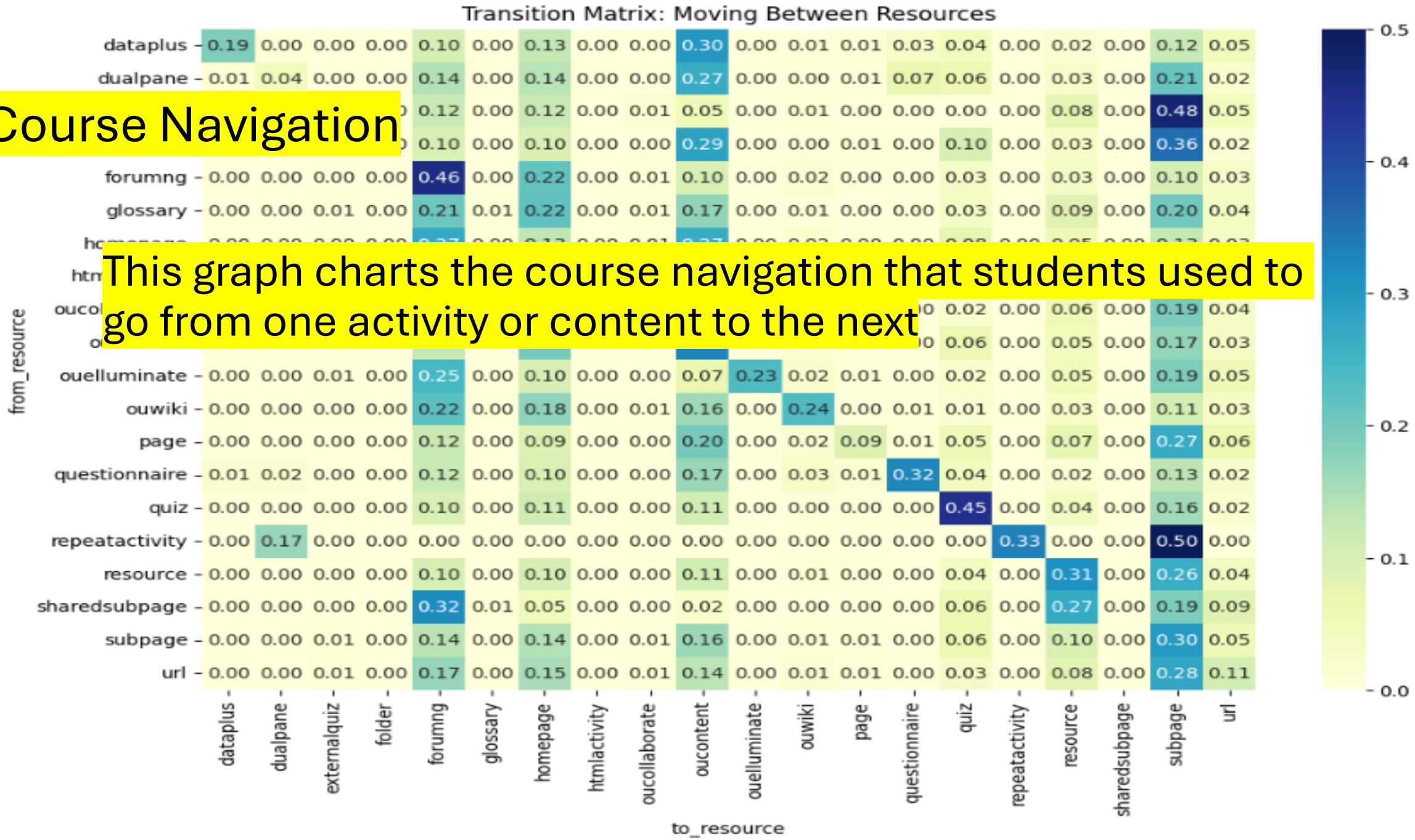
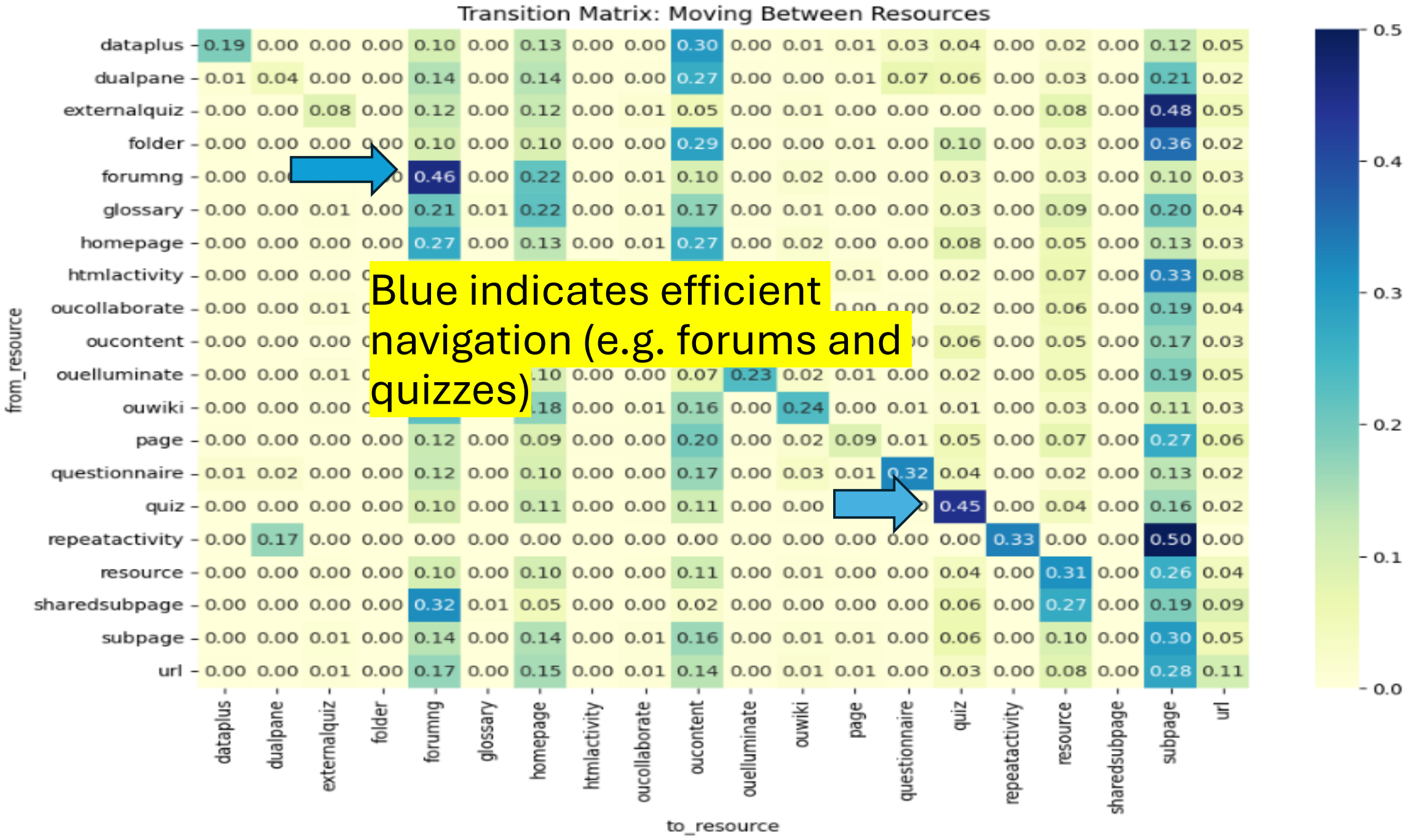

Average Social Activity per Performance Group
Connection is key
Average Posts
Average Reads
0
1
2
3
4
5
6
0
20
40
60
80
100
120
Fail
Pass
Withdrawn
Final Result

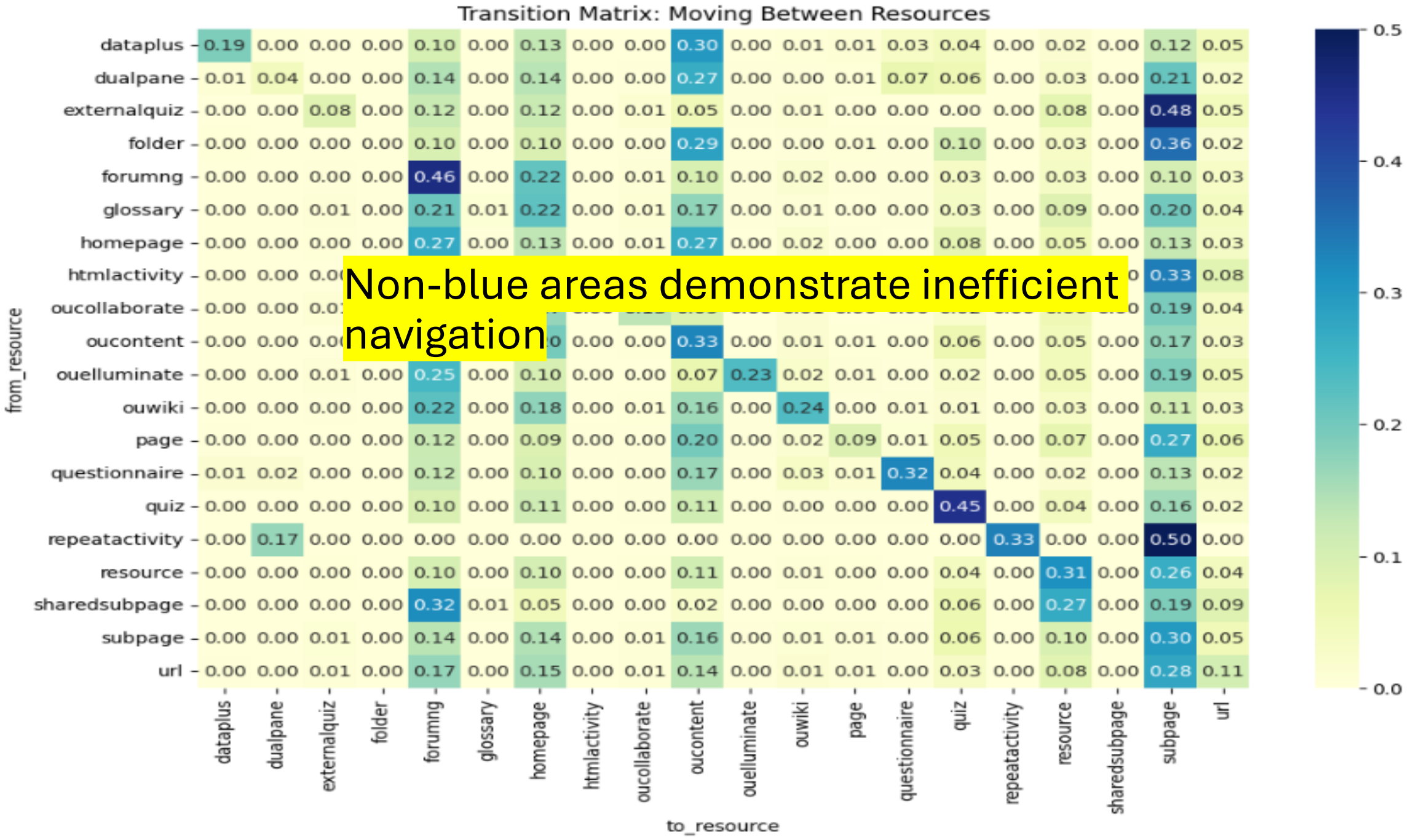

Average Social Activity per Performance Group
Posting and Reading were key indicators of success
Average Posts
Average Reads
0
1
2
3
4
5
6
0
20
40
60
80
100
120
Fail
Pass
Withdrawn
Final Result

Course Navigation

This graph charts the course navigation that students used to go from one activity or content to the next

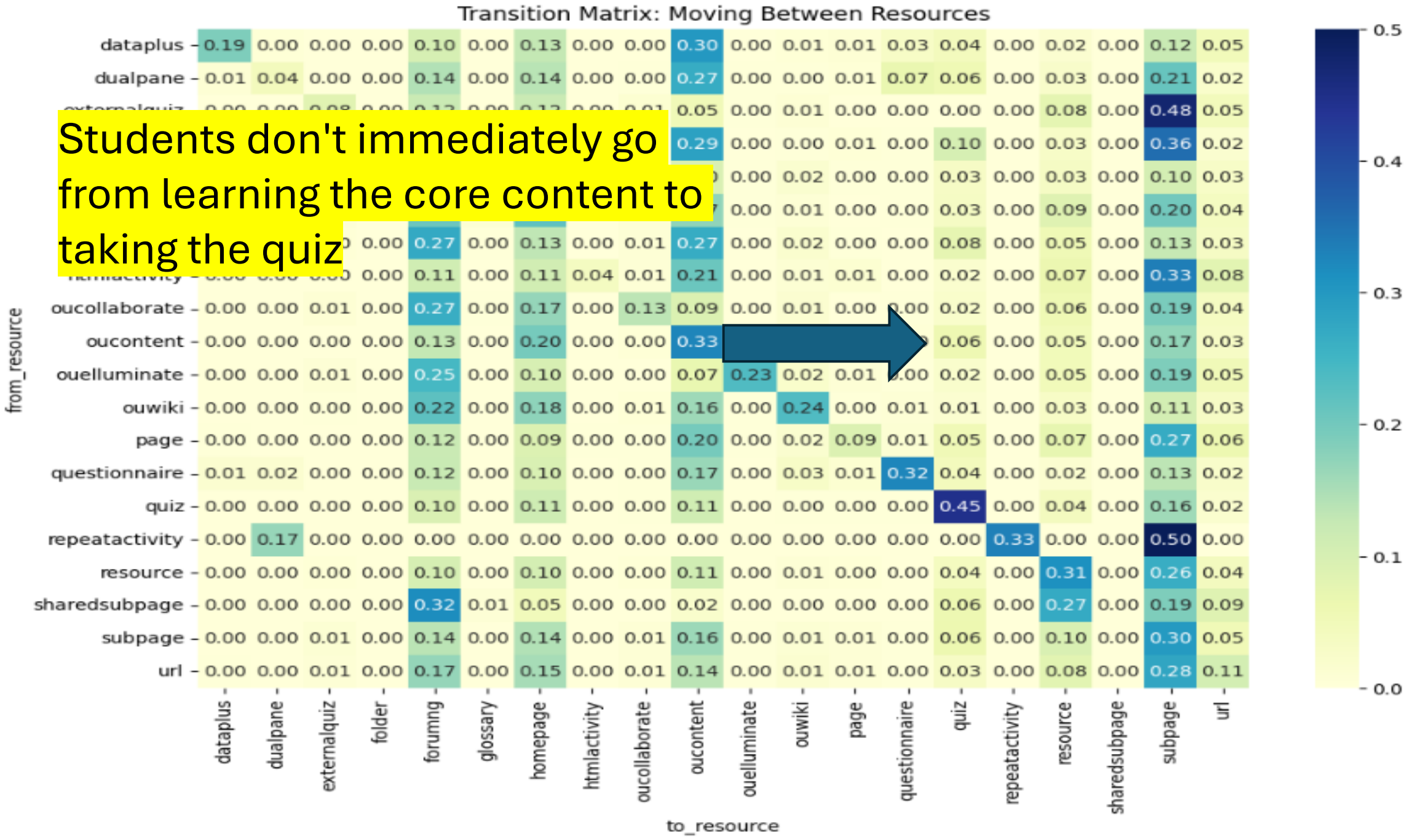


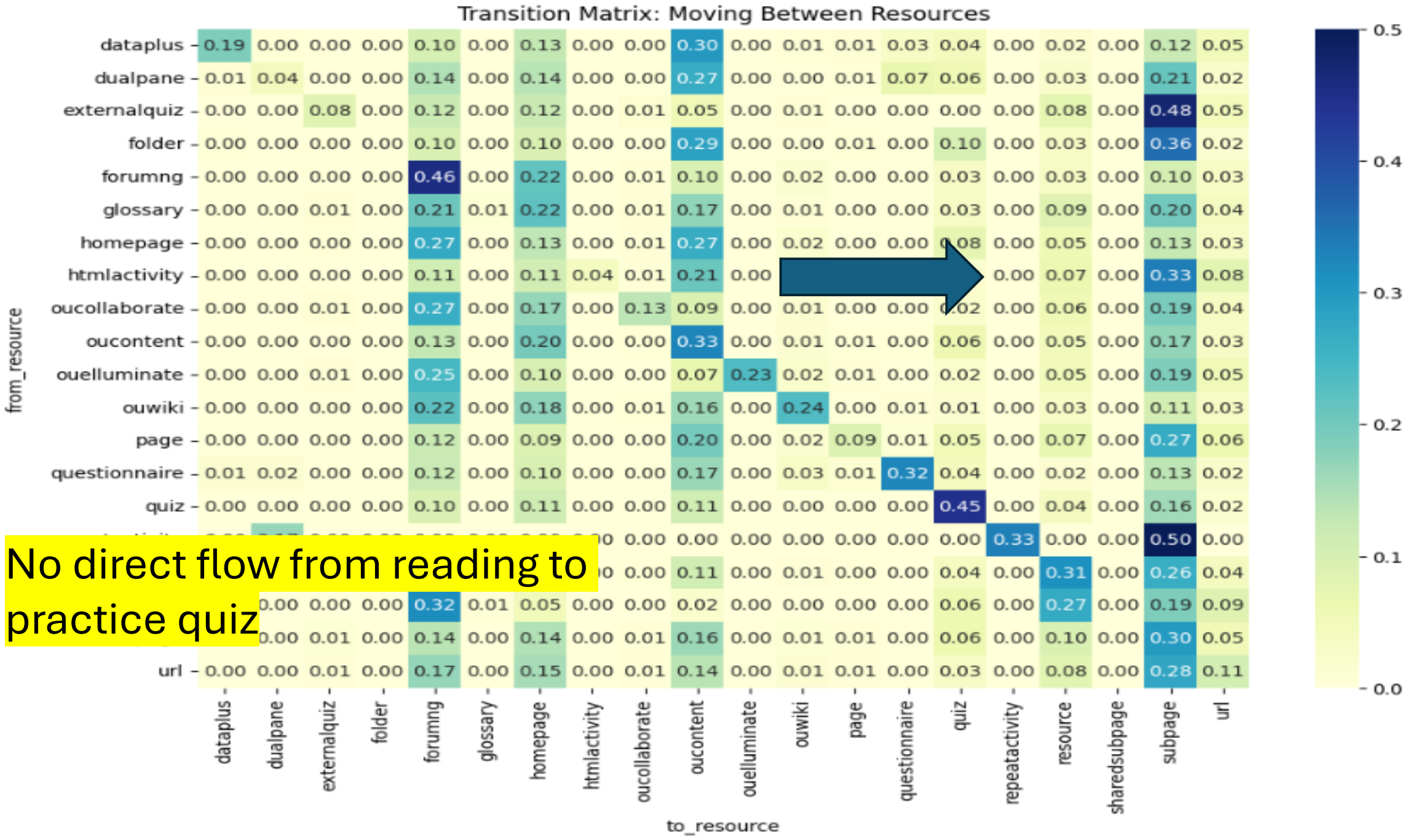

Transition Matrix: Moving Between Resources
from_resource
to_resource
dataplus
dualpane
externalquiz
folder
forumng
glossary
homepage
htmlactivity
oucollaborate
oucontent
ouelluminate
ouwiki
page
questionnaire
quiz
repeatactivity
resource
sharedsubpage
subpage
url
0.0
0.1
0.2
0.3
0.4
0.5
Blue indicates efficient navigation (e.g. forums and quizzes)



Non-blue areas demonstrate inefficient navigation

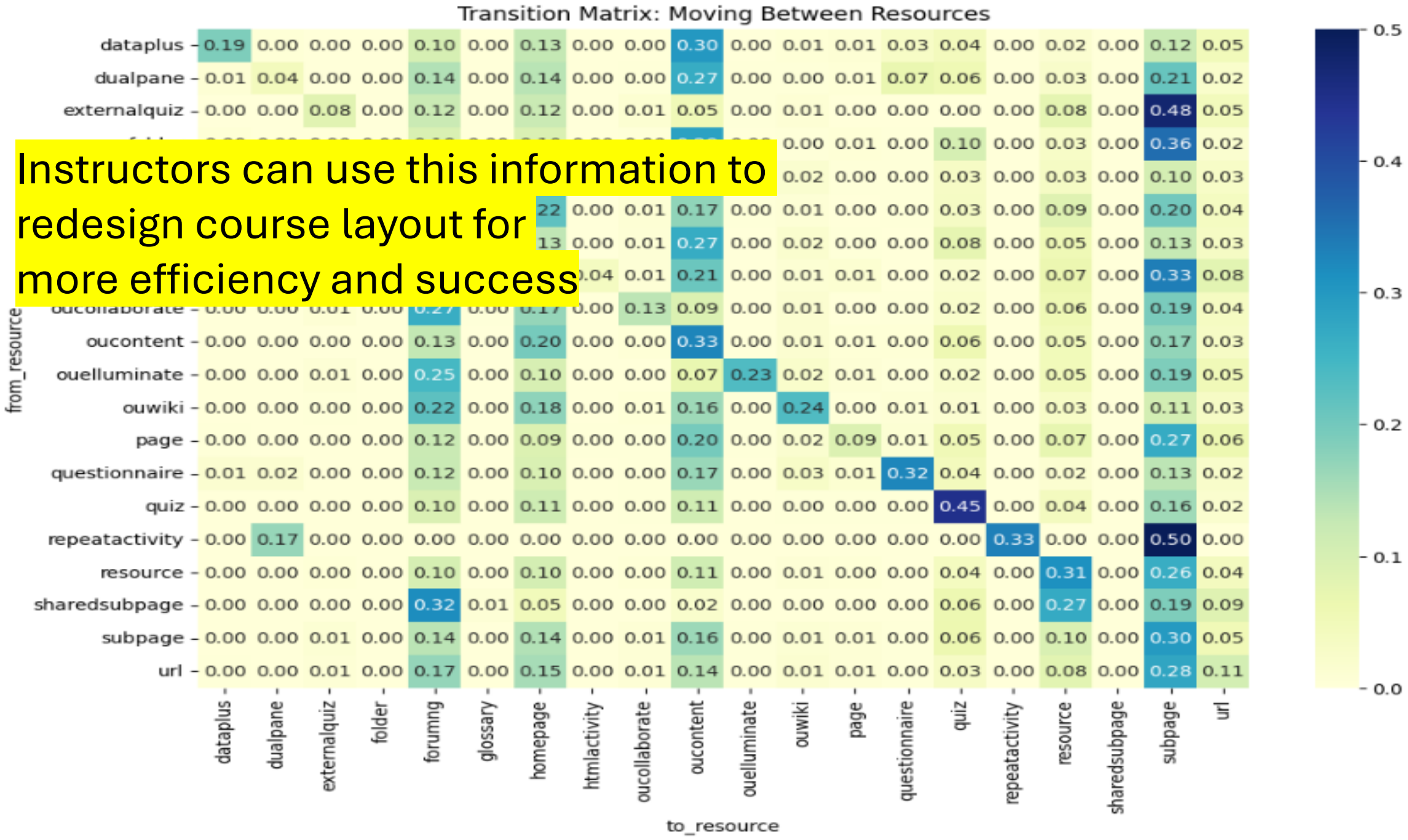
Students don't immediately go from learning the core content to taking the quiz





No direct flow from reading to practice quiz

Instructors can use this information to redesign course layout for more efficiency and success



From Analysis to Prediction

- Use early-course data from the first 20% of the course
- Predict student outcomes before the course ends
- Enable early identification of at-risk students

Machine Learning Approach

- Learn patterns from past students
- Apply patterns to new students
- Compare multiple models
- Select best-performing method

Prediction Tasks

- Multiclass classification
  - Distinction
  - Pass
  - Fail
  - Withdrawn
- Binary classification
  - At-risk
  - Not at-risk

Features Used for Prediction

- Early engagement
  - Total clicks
  - Active days
- Early assessment performance
  - Average score
  - Max score
- Student information
  - Education
  - Age
  - Region
- Registration behavior

Models Tested

-Logistic Regression

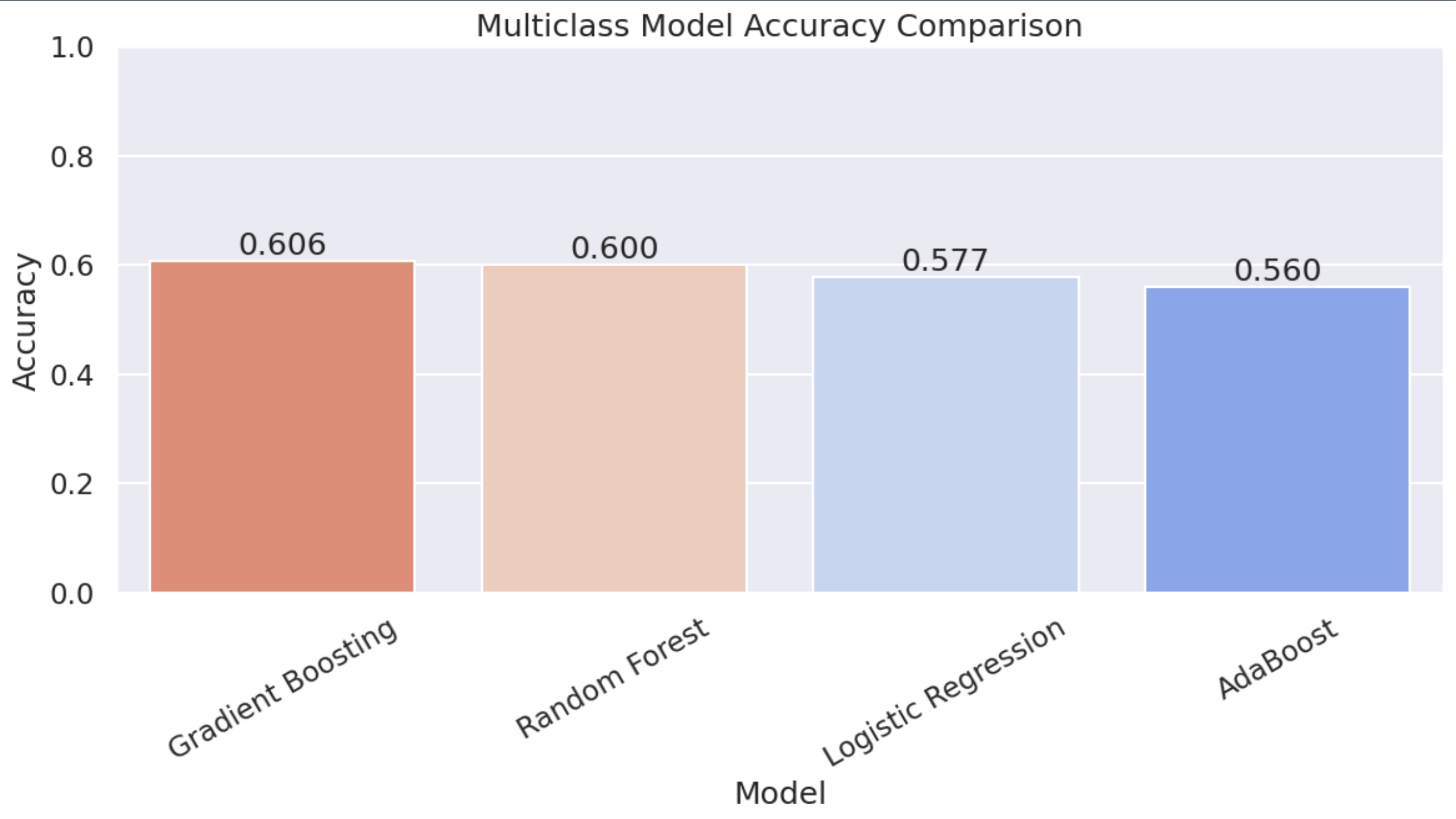
-Random Forest

-Gradient Boosting

-AdaBoost

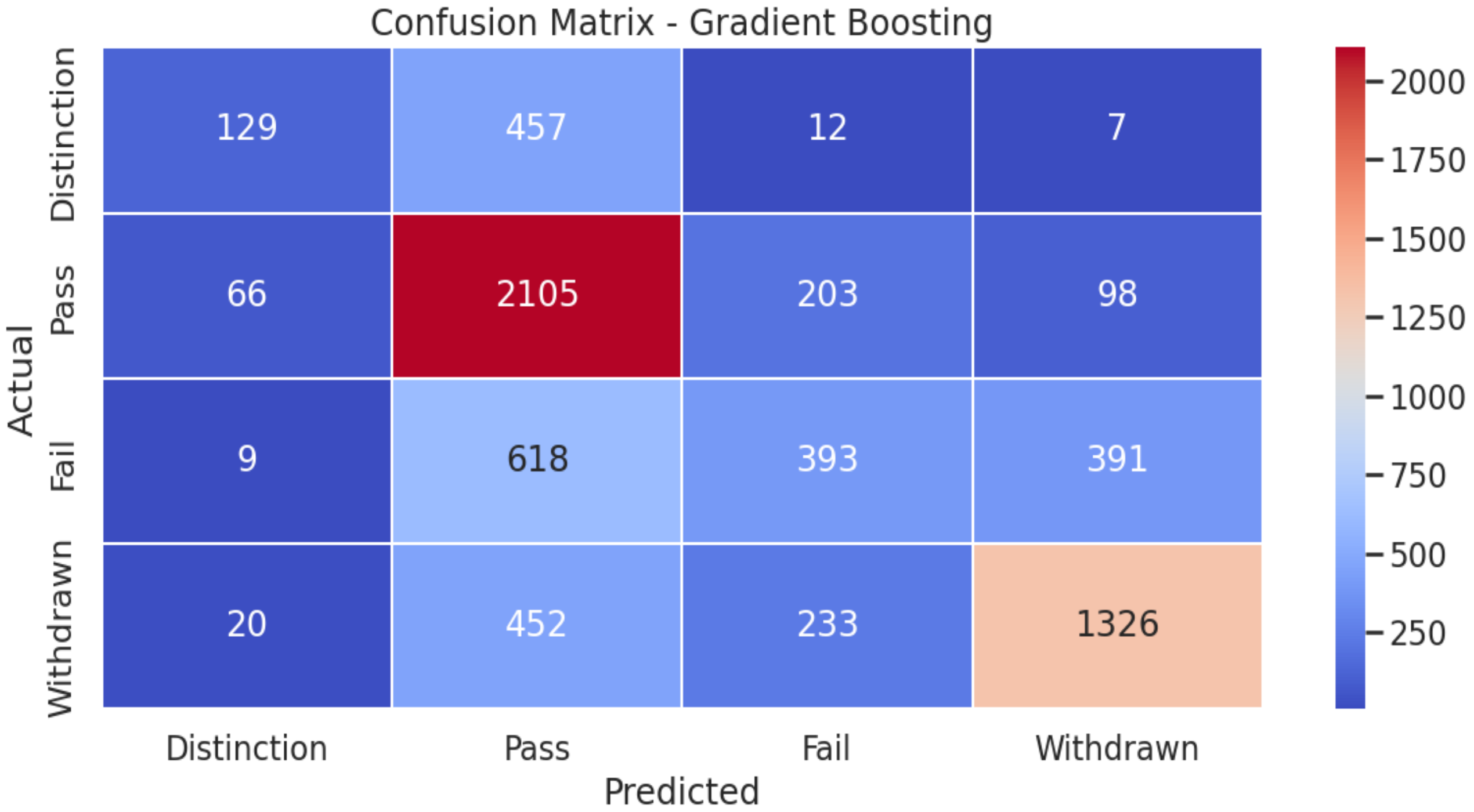
Multiclass Prediction Results

- Compare multiple machine learning models
- Evaluate performance
- Show highest accuracy model


Multiclass Model Accuracy Comparison
1.0
0.8
0.6
0.4
0.2
0.0
Accuracy
0.606
0.600
0.577
0.560
Gradient Boosting
Random Forest
Logistic Regression
AdaBoost
Model

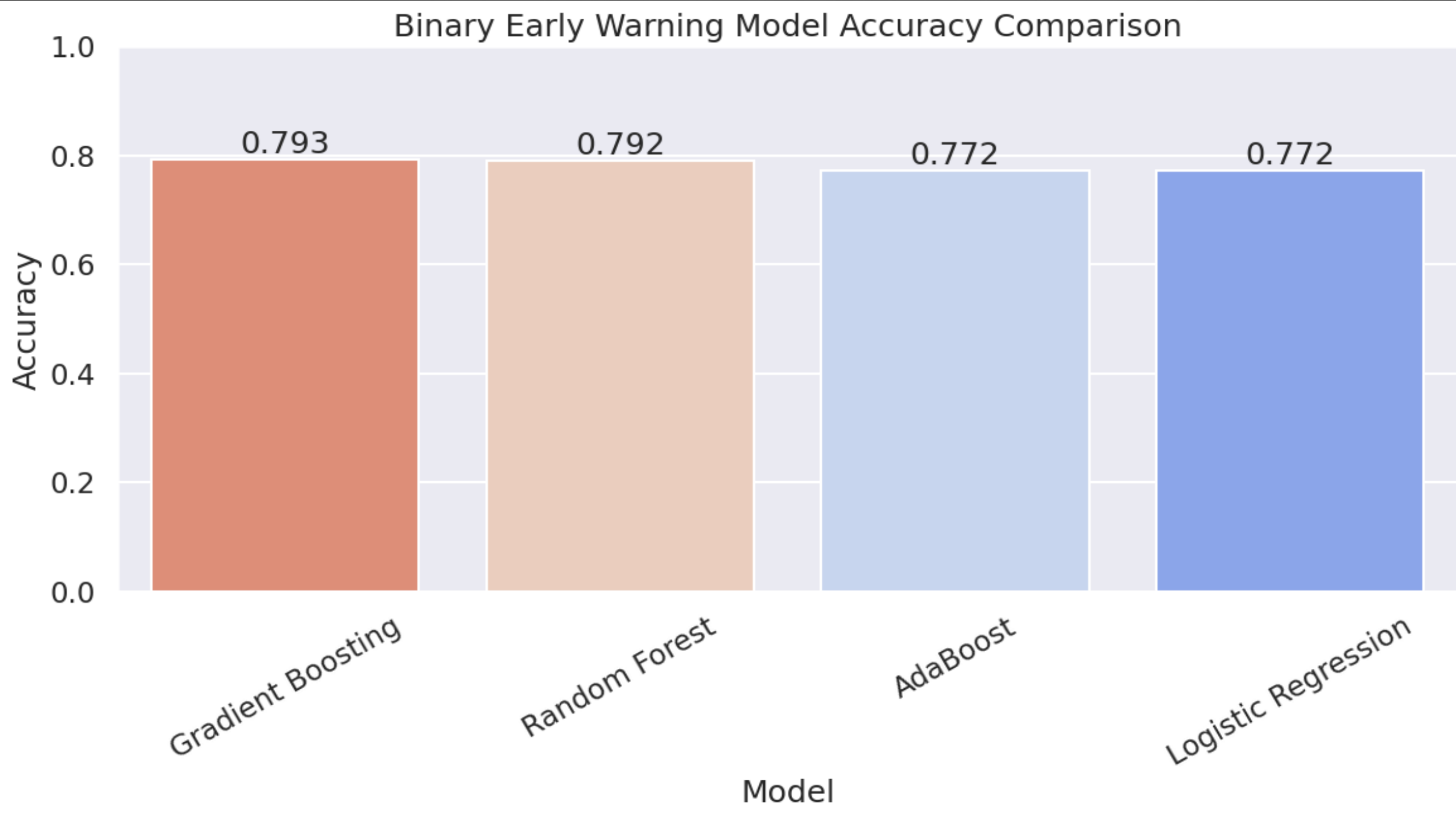
Multiclass Confusion Matrix

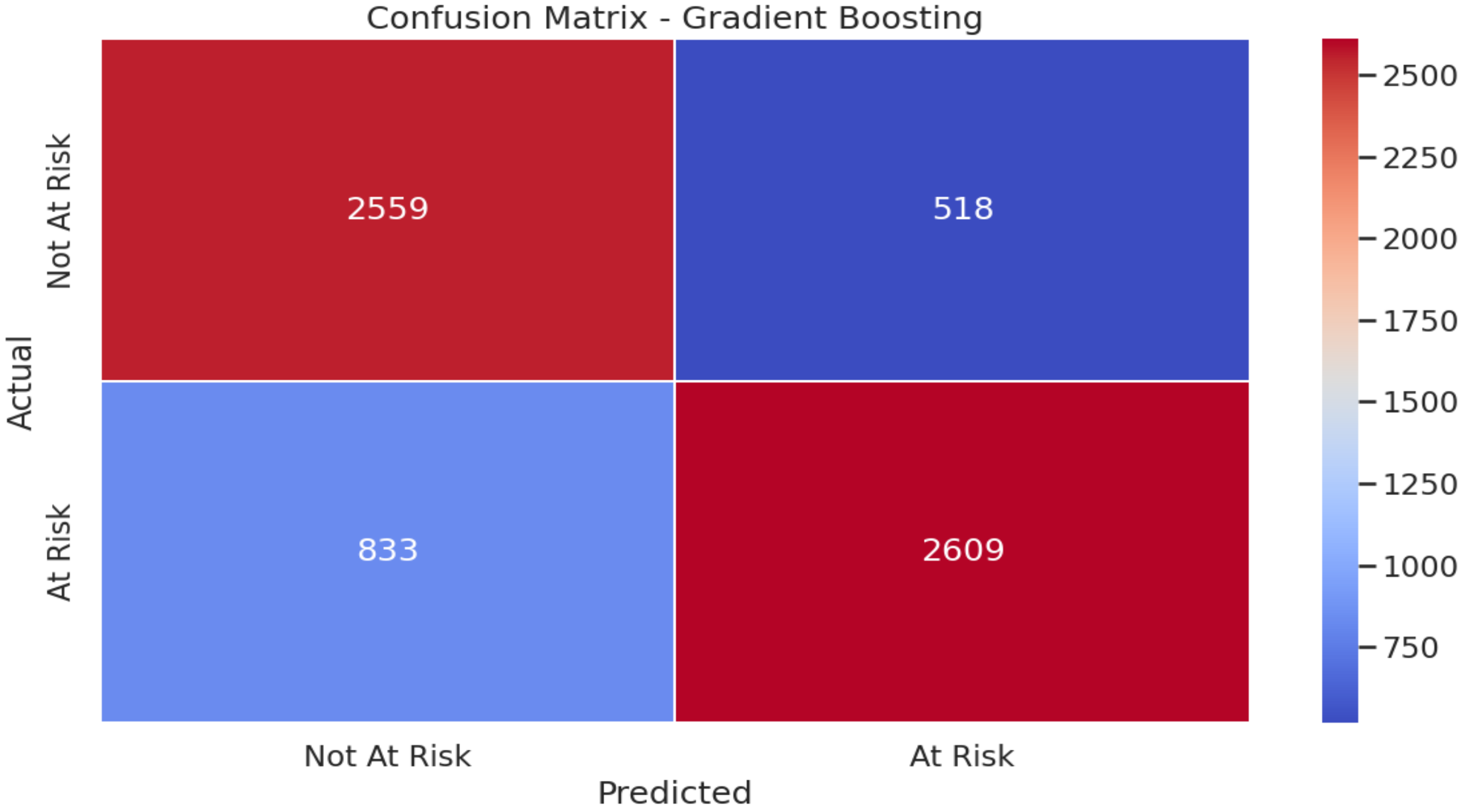
- Breaks predictions down by class
- Shows correct vs. incorrect classifications
- Helps visualize where the model is making mistake


Confusion Matrix - Gradient Boosting
Actual
Distinction
Pass
Fail
Withdrawn
129
457
12
7
66
2105
203
98
9
618
393
391
20
452
233
1326
Distinction
Pass
Fail
Withdrawn
Predicted
2000
1750
1500
1250
1000
750
500
250

Binary Classification Results

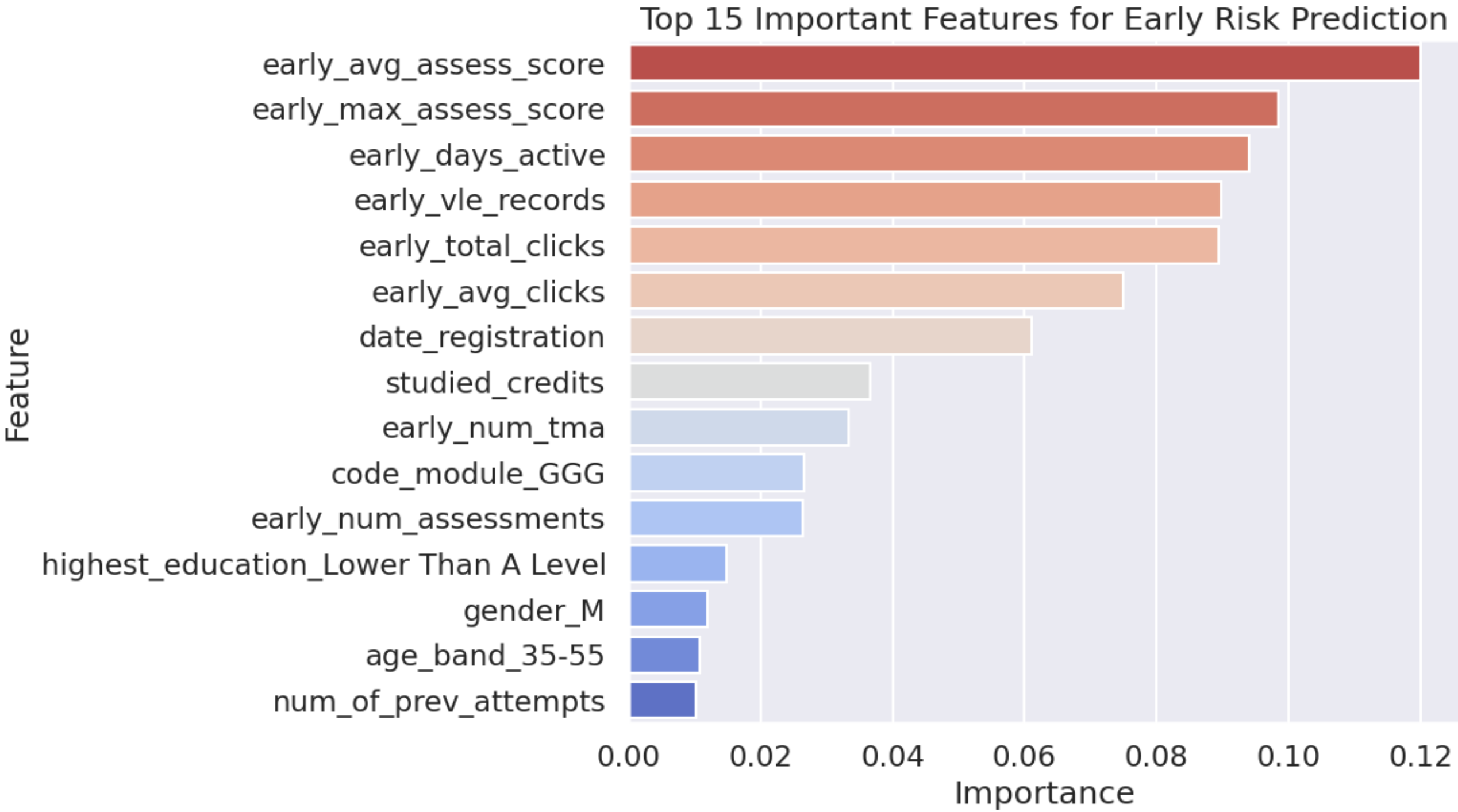
- Binary framing removes unnecessary categories
- Highlights students who may fail or withdraw
- Makes the model’s decisions simpler to interpret and act on


Binary Early Warning Model Accuracy Comparison
1.0
0.8
0.6
0.4
0.2
0.0
Accuracy
0.793
0.792
0.772
0.772
Gradient Boosting
Random Forest
AdaBoost
Logistic Regression
Model


Confusion Matrix - Gradient Boosting
Actual
Not At Risk
At Risk
2559
518
833
2609
Not At Risk
At Risk
Predicted
2500
2250
2000
1750
1500
1250
1000
750

Feature Importance

- What drives predictions
- Key features
- Explain model results
- Insights from variables


Top 15 Important Features for Early Risk Prediction
early_avg_assess_score
early_max_assess_score
early_days_active
early_vle_records
early_total_clicks
early_avg_clicks
date_registration
studied_credits
early_num_tma
code_module_GGG
early_num_assessments
highest_education_Lower Than A Level
gender_M
age_band_35-55
num_of_prev_attempts
Feature
0.00
0.02
0.04
0.06
0.08
0.10
0.12
Importance

Causal Effects: From Correlation to Causation

Correlation: Students who click more tend to succeed

Limitation: Correlation does not imply causation

Key Question:

- Does engagement *cause* success?
- What happens if we increase engagement?

- Approach:
  - Compare similar students
  - Control for background factors

Causal Effect of Engagement on Success

Treatment: High engagement (high clicks)

Outcome: Passing the course

Result (Average Treatment Effect (ATE)):

- **+32.3% increase in pass probability**

Interpretation:

- Increasing engagement improves success

Implication:

- Early intervention strategies can be effective

Final Machine Learning Insight

- Early behavior predicts later outcomes
- Engagement has a causal effect on success
- Engagement and assessment matter most
- Binary prediction is practical and useful
- Machine learning can support intervention

Sources

Kuzilek, J., Hlosta, M. & Zdrahal, Z. (2017) Open University Learning Analytics dataset. *Sci Data,* 4, 170171

https://doi.org/10.1038/sdata.2017.171

The Open University. (n.d.). *Facts and figures*. https://www.open.ac.uk/policies-and-reports/facts-and-figures *(Accessed April 13, 2026)*

Sim, S. (2024). *Analysis of the OULAD dataset*. GitHub Pages. https://sergiosim.github.io/OULAD/

Thank you!

Questions?